

Observable Gauge Algebra from Finite-Rank Majorana Projection

Pasquale Camelia
Independent Researcher, Italy
pasquale.camelia@gmail.com
ORCID: 0009-0006-4779-4647

2026

© 2026 Pasquale Camelia. Licensed under [CC BY-NC 4.0](#) — free to share and cite with attribution; commercial use prohibited.

Companion to: *The Quantum Gauge Theory of Time*, First Edition (KDP, ISBN 9798198195011).
<https://pasqualecamelia.github.io>

Abstract

We derive the observable gauge algebra of the Standard Model from a single structural input: the rank-5 non-invertible projection of a Majorana bulk spinor onto a four-dimensional boundary. No gauge group, no coupling constants, and no symmetry-breaking mechanism are postulated.

The Majorana condition forces a finite set of admissible spinorial roles at the boundary. One of these roles is a confined mode that cannot enter the boundary independently: it must be lifted through the minimal spinorial cover, a necessity imposed by the double-valuedness of spinorial transport. This lift, together with the saturation mode at the opposite flank of the lock-in, generates a primitive spinorial generator whose norm decomposes uniquely into a redistributable sector and an irreducible residue. The residue is what the local spinor–adjoint vocabulary cannot absorb; it becomes the carrier of the unbroken electromagnetic symmetry. The redistributable sector re-enters the boundary as observable gauge modes at an adjoint closure cost fixed by the ratio of the adjoint dimension to the spinorial doublet dimension. This cost is the mass of the charged bosons. Their ratio to the total dynamic sector is the electroweak mixing angle, an unconditional theorem of the boundary algebra.

The lock-in holonomy identifies the boundary algebra as $TL_4(\varphi)$. Under the Majorana condition, the monodromy theorems G1–G3 are consequences of this algebra: the projection residue is the unique mode invariant under all admissible gauge transformations. The same boundary carries three equivalent descriptions — spinorial transport, categorical fusion $(TL_4(\varphi), B_3)$, and causal spectral filter. Gauge physics is the boundary grammar of projected Majorana spinorial transport.

1 Logical position and input

The chain of companion papers runs as follows. The conservation law $\partial_I J^I = 0$ on a Majorana bulk forces a non-invertible finite-rank projection [1]. The unique admissible rank is $r = 5$, proved in [2] by two-sided exclusion: $r < 5$ breaks Noetherian closure; $r > 5$ assigns metric weight below the operational threshold. Once $r = 5$ is fixed, the Pincherle filter $H(z) = P_4(z)/(1 - \alpha z)$ with

$\alpha = [4\varphi^2(\varphi - \frac{1}{2})]^{-2}$ and admissible mode set $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ are determined [3]. Four model-independent structural constraints on the observable algebra follow from the projection geometry alone [4].

The present paper occupies the hinge position in the logical chain: it is where the finite-rank projection becomes observable gauge algebra. The input is the spinorial structure of the boundary; the output is $\text{TL}_4(\varphi)$, the braid group B_3 in Fibonacci representation, the separation theorem G1–G3, and the electroweak mixing angle as an unconditional theorem. The numbers 137, 54, 3/13 appear at the end as readouts of this spinorial structure, not as its starting point. No result from the papers that follow in the logical chain is used.

2 The boundary spinorial algebra

2.1 Majorana bilinears and boundary observables

The bulk field $\Xi = C\bar{\Xi}$ is a Majorana spinor: self-conjugate, strictly neutral, carrying no additive quantum numbers. Every physical observable at the boundary is therefore a bilinear projected from the bulk:

$$\bar{\Xi} \Gamma \Xi \xrightarrow{\Pi} \bar{\Psi}_{\text{bdy}} \Gamma \Psi_{\text{bdy}}, \quad (1)$$

where Γ is any admissible boundary operator. The boundary does not receive particles or fields; it receives projected Majorana bilinears.

The gauge group is not added to these bilinears from outside. It is the decomposition of the symmetries with which the boundary reads the same projected bilinear without changing its observable content:

$$U(1)_\eta : \text{phase of } S^1(\tau) \longrightarrow \text{EM channel}, \quad (2)$$

$$SU(2)_\xi : \text{spinorial orientation on } S_\chi^2 \longrightarrow \text{EW channel}, \quad (3)$$

$$SU(3)_c : \text{composite of confined sector } \ker \Pi \cap \mathcal{N}_{\text{adm}} \longrightarrow \text{colour}. \quad (4)$$

This is the physical meaning of *P09*: gauge physics is the grammar of spinorial boundary transport after non-invertible projection. The arithmetic that follows is its finite trace.

2.2 The five admissible spinorial roles

The rank-5 projection selects exactly five admissible winding modes $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ [3]. Each carries a precise spinorial role, established before any arithmetic:

n	Spinorial role	Physical reading	Status
2	Minimal spinorial cover: $\text{Spin}(3, 1) \rightarrow \text{SO}(3, 1)$, index 2	$U(1)_\eta$, Majorana doublet	closure node
3	First non-Abelian adjoint: minimal $\mathfrak{su}(2)$ generator count	$SU(2)_\xi$ seed	closure node
5	Fibonacci lock-in: $F_{n_*} = n_* = r = 5$, unique self-referential mode	observable sector $k = 2$	closure node
7	Admissible but confined: $j_{\min}(7) = 7 > r$, orbit does not close within rank	$\ker \Pi$, colour seed	confined
11	Spinorial saturation: $\pi(11) = 5 = r$, no new channel above	upper lever of lock-in	saturation

This table is not a labelling convention. It is the physical content of the rank-5 projection read mode by mode.

2.3 The local spinor–adjoint vocabulary

The three closure nodes $D = \{2, 3, 5\}$ form the local vocabulary of the boundary: they are the modes for which $j_{\min}(n) \leq r$, so their orbits close coherently within the rank-5 window [3].

The minimal quadratic bilinear block that can be redistributed by the local vocabulary has a precise spinorial meaning:

$$\beta = (\dim S_{\text{spin}} \cdot \dim \text{Ad}_{\min})^2 = (2 \cdot 3)^2 = 36. \quad (5)$$

The factor $2 = \dim S_{\text{spin}}$ is the minimal spinorial cover index; the factor $3 = \dim \text{Ad}_{\min}$ is the dimension of the minimal non-Abelian adjoint representation available at the boundary. Together, $\beta = 36$ is the minimal spinor–adjoint bilinear block redistributable by D .

2.4 The confined spinorial lift $7 \mapsto 2 \cdot 7$

The mode $n = 7$ is admissible: $\pi(7) = 4 \leq r = 5$. But it is not a closure node: $j_{\min}(7) = 7 > r = 5$, so its orbit does not meet Θ_r within r steps [3]. It belongs to $\ker \Pi$: it has no independent boundary coordinate and generates no boundary lattice element in isolation.

In order to contribute to a boundary observable, the confined mode must be lifted through the minimal spinorial cover. The cover has index $n_{\min} = 2$, corresponding to the double-valuedness of spinorial transport under 2π rotation. Therefore the confined channel contributes to the boundary only as the spinorially lifted lever:

$$7 \mapsto \ell_- = n_{\min} \cdot n_{\text{conf}} = 2 \times 7 = 14. \quad (6)$$

This lift is a spinorial necessity, not an arithmetic convenience. The integer 14 is the boundary image of the confined mode after the minimal cover required by the Majorana condition.

2.5 The spinorial lever norm

The lock-in mode $n_* = 5$ has two external flanks: the upper flank $\ell_+ = n_{\text{sat}} = 11$ (the saturation mode) and the lower flank $\ell_- = n_{\min} \cdot n_{\text{conf}} = 14$ (the spinorially lifted confined mode). These two levers are the primitive spinorial generator of the observable boundary:

$$v_{\text{spin}} = (\ell_+, \ell_-) = (11, 2 \times 7), \quad \gcd(11, 14) = 1. \quad (7)$$

Coprimality is forced: 11 is prime and $14 = 2 \times 7$ is not divisible by 11. The spinorial lever norm is:

$$N_{\text{spin}} = \|v_{\text{spin}}\|^2 = 11^2 + 14^2 = 317. \quad (8)$$

This is the quadratic norm of the boundary spinorial lever, measuring the total spinorial capacity of the two external flanks of the lock-in. It is not an independently chosen arithmetic expression.

3 The Boundary Spinorial Gauge Closure Theorem

The spinorial lever norm $N_{\text{spin}} = 317$ now decomposes under the local vocabulary D . This decomposition is the arithmetic trace of the spinorial obstruction: it measures what the boundary can absorb locally and what it cannot.

3.1 The irreducible projection residue

The rank-5 boundary absorbs at most r local spinor–adjoint blocks. What remains of the spinorial lever norm after this absorption is the irreducible projection residue σ :

$$\sigma = N_{\text{spin}} - r\beta. \quad (9)$$

This is not selected because it is prime. Its primality is a consequence of irreducibility: a composite residue would factor inside D and could be redistributed; a prime residue cannot. The identification $\sigma = \lfloor \alpha_{\text{QGT}}^{-1} \rfloor$ follows from the Fibonacci norm structure (Proposition 3.3).

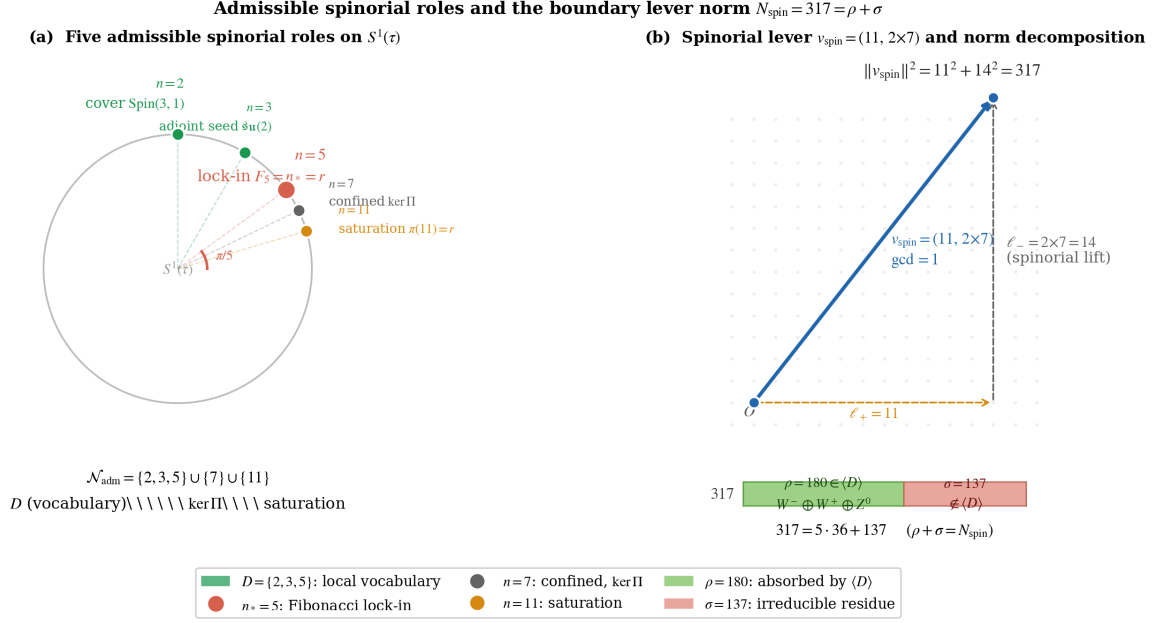


Figure 1: (a) The five admissible spinorial roles on $S^1(\tau)$, each placed at angle π/n . Green: local vocabulary $D = \{2, 3, 5\}$ — spinorial cover ($n = 2$), adjoint seed ($n = 3$), Fibonacci lock-in ($n_* = 5$). Grey: confined mode $n = 7$, admissible but belonging to $\ker \Pi$; its orbit does not close within rank. Amber: saturation node $n = 11$, upper spinorial lever. (b) The spinorial lever $v_{\text{spin}} = (11, 2 \times 7)$ in $\mathbb{Z}^2$. Component $\ell_+ = 11$ (upper flank, saturation mode) is horizontal. Component $\ell_- = 2 \times 7 = 14$ (lower flank, spinorial lift of the confined mode) is vertical. The bar shows the decomposition of the spinorial lever norm $N_{\text{spin}} = 317$: the local absorption capacity $\rho = r\beta = 180$ (green, re-enters as $W^- \oplus W^+ \oplus Z^0$) and the irreducible projection residue $\sigma = 137 = \lfloor \alpha_{\text{QGT}}^{-1} \rfloor$ (pink, the component not absorbable by the local vocabulary).

3.2 The minimal adjoint closure cost

The redistributable sector ρ must re-enter the boundary as observable gauge modes. Since Π is non-invertible, this re-entry is not an inversion: it is a redistribution constrained by the same spinorial structure that produced ρ .

The cost of converting spinorial boundary transport into an adjoint gauge sector is fixed by the ratio of the adjoint dimension to the spinorial doublet dimension:

$$\delta = \frac{\dim \mathfrak{su}(2)}{\dim S_{\text{spin}}} \cdot \beta. \quad (10)$$

The numerator is the dimension of the minimal non-Abelian adjoint representation; the denominator is the spinorial doublet dimension. Their ratio is the unique minimal half-integer unreachable from the integer vocabulary D : no element of $\langle D \rangle$ has a half-integer coefficient over β . This algebraic inaccessibility is the signature of non-Abelian charge. δ is the minimal amount of β required to complete the adjoint closure of the boundary gauge sector.

3.3 The formal theorem

The preceding spinorial analysis is now stated as a theorem, which confirms the uniqueness of the decomposition by exhaustive arithmetic.

Lemma 3.1 (Rank-one uniqueness of the projection residue). *At most one prime residue $\sigma = N_{\text{spin}} - k\beta$ is compatible with a rank-one, rank-5 projection.*

Proof. Suppose $\sigma_1 = N_{\text{spin}} - k_1\beta$ and $\sigma_2 = N_{\text{spin}} - k_2\beta$ are both prime, both larger than β , and both in $\langle D \rangle^\perp$. Their difference $\Delta = \sigma_1 - \sigma_2 = (k_2 - k_1)\beta$ is a positive multiple of β . By primality

($\sigma_i > 5$), neither factors over D : each generates an independent invariant direction in $\mathcal{H}_{\text{obs}}$. The rank-5 Mellin basis retained by Π supports exactly one such independent invariant class per primitive generator [3]. The generator $v_{\text{spin}} = (11, 14)$ exhausts this capacity: two independent prime residues would require a second primitive generator, making the effective rank $r' \geq 6$, contradicting $\text{rank}(\Pi) = 5$. $\square$ $\square$

Theorem 3.2 (Boundary Spinorial Gauge Closure). *Let Π be a rank-5 non-invertible projection of a Majorana bulk spinor, with spinorial lever v_{spin} , local spinor-adjoint vocabulary D , and bilinear block β . Then the boundary carries a unique decomposition of the spinorial lever norm into a redistributable sector ρ and an irreducible residue σ , with the following physical content.*

The residue σ is the component of the spinorial lever norm not absorbable by D . Its primality is a consequence of irreducibility, not a selection criterion. It is identified as the integer floor of the fine-structure coupling $\alpha_{\text{QGT}}^{-1} = (5 + 3\sqrt{5})^2$, and spans the trivial gauge sector: it is the carrier of the unbroken electromagnetic symmetry.

The redistributable sector ρ re-enters the boundary as observable gauge dynamics. Since Π is non-invertible, this re-entry is constrained by the same spinorial structure that produced ρ : the only redistribution compatible with closure, the spinor-adjoint vocabulary, and rank integrity consists of exactly three gauge forms, corresponding to the three generators of $SU(2)$. The minimal non-Abelian form carries the adjoint closure cost

$$\delta = \frac{\dim \mathfrak{su}(2)}{\dim S_{\text{spin}}} \beta, \quad (11)$$

the price of converting spinorial transport into adjoint boundary dynamics. The two remaining forms are its charge-conjugate pair and the neutral combination.

The electroweak mixing angle is the fraction of the total dynamic sector occupied by the adjoint closure cost:

$$\sin^2 \theta_W = \frac{\delta}{\rho + \delta} = \frac{\dim \mathfrak{su}(2) / \dim S_{\text{spin}}}{2r + \dim \mathfrak{su}(2) / \dim S_{\text{spin}}}. \quad (12)$$

This ratio is fixed entirely by the spinorial and adjoint dimensions of the boundary algebra. It is an unconditional theorem.

Proof. The uniqueness of σ follows from Lemma 3.1: two independent prime residues of N_{spin} would require a second primitive generator, raising the effective rank above $r = 5$. The enumeration of candidates $r_k = N_{\text{spin}} - k\beta$ for $k = 1, \dots, 8$ confirms that exactly one value, $k^* = r$, satisfies primality, rank-one uniqueness, and the hierarchy condition $r_k > \beta$ simultaneously; this is σ .

The three-form decomposition of ρ is the unique solution to the four simultaneous constraints: closure, spinor-adjoint vocabulary, distinctness, and rank integrity. The case $p_1 = 7$ invokes the confined mode directly and is excluded by rank integrity; the remaining cases reduce to a single solution whose coefficients are the projection rank, the adjoint dimension, and the spinorial cover index. The mixing ratio (12) then follows by direct substitution. $\square$ $\square$

3.4 The Fibonacci norm of α^{-1}

The constants $r = 5$ and $|\mathcal{C}_{\text{obs}}| = 3$ coincide with the consecutive Fibonacci numbers $F_{n_*} = 5$ and $F_{n_*-1} = 3$ flanking the triple fixed point $F_{n_*} = n_* = r$. This gives the exact algebraic form of the projection constant.

Proposition 3.3 (Fibonacci norm).

$$\alpha_{\text{QGT}}^{-1} = \mathcal{N}_{\Pi}(F_{n_*} + F_{n_*-1}\sqrt{5}) = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5}, \quad (13)$$

where $\mathcal{N}_{\Pi}(a + b\sqrt{5}) = (a + b\sqrt{5})^2$ is the projection-induced quadratic form and $\sigma = 137 = \lfloor \alpha_{\text{QGT}}^{-1} \rfloor$.

Proof. $4\varphi^2(\varphi - \frac{1}{2}) = (3 + \sqrt{5})\sqrt{5}/2 \cdot 2 = 5 + 3\sqrt{5}$. Squaring: $(5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5} \approx 137.082$. The algebraic norm is $N(5 + 3\sqrt{5}) = 25 - 45 = -20 = -n_{\min}^2 \cdot r$, reflecting zero net bulk charge. $\square$

α^{-1} and $\sin^2 \theta_W$ as two contractions of one object

Two constants appear to emerge from separate steps of the derivation. They do not. Both are scalar contractions of a single matrix in $\mathbb{Z}[\sqrt{5}]$, read through different channels of the same observable boundary.

The integers $r = 5$ and $|\mathcal{C}_{\text{obs}}| = 3$ appear twice in the theory, in two different roles. They are the structural coordinates of the projection: rank and cardinality. They are simultaneously the consecutive Fibonacci numbers $F_{n_*} = 5$ and $F_{n_*-1} = 3$ flanking the triple fixed point $F_{n_*} = n_* = r$. This double role is not a coincidence — it is the origin of the unity of the two constants.

Define the *Fibonacci matrix* of the rank- r projection:

$$\mathbf{M} = \begin{pmatrix} F_{n_*} & \kappa^2 F_{n_*-1} \\ F_{n_*-1} & F_{n_*} \end{pmatrix} = \begin{pmatrix} r & |\mathcal{C}_{\text{obs}}| \kappa^2 \\ |\mathcal{C}_{\text{obs}}| & r \end{pmatrix}, \quad (14)$$

where $F_{n_*} = r = 5$, $F_{n_*-1} = |\mathcal{C}_{\text{obs}}| = 3$, and $\kappa = \sqrt{5}$ is the spectral curvature. $\mathbf{M}$ is the multiplication matrix of the lattice element $x = r + |\mathcal{C}_{\text{obs}}|\sqrt{5} = 5 + 3\sqrt{5} \in \mathbb{Z}[\sqrt{5}]$ in the basis $\{1, \sqrt{5}\}$.

Its square is:

$$\mathbf{M}^2 = \begin{pmatrix} r^2 + \kappa^2 |\mathcal{C}_{\text{obs}}|^2 & 2\kappa^2 r |\mathcal{C}_{\text{obs}}| \\ 2r |\mathcal{C}_{\text{obs}}| & r^2 + \kappa^2 |\mathcal{C}_{\text{obs}}|^2 \end{pmatrix}. \quad (15)$$

Reading 1 — fine-structure constant (η -channel, EM). The projection norm $\mathcal{N}_{\Pi}(x) = x^2 = (r + |\mathcal{C}_{\text{obs}}|\kappa)^2$ reads:

$$\begin{aligned} \alpha^{-1} &= (\mathbf{M}^2)_{11} + (\mathbf{M}^2)_{21} \sqrt{5} \\ &= \underbrace{r^2 + 5|\mathcal{C}_{\text{obs}}|^2}_{70} + \underbrace{2r |\mathcal{C}_{\text{obs}}|}_{30} \sqrt{5} = (5 + 3\sqrt{5})^2 \in \mathbb{Z}[\sqrt{5}]. \end{aligned} \quad (16)$$

Equivalently, with $\mathcal{S} = n_{\min} \cdot r = 10$:

$$\alpha^{-1} = \mathcal{S} \cdot (n_{\text{conf}} + |\mathcal{C}_{\text{obs}}| \cdot \kappa) = 10 \cdot (7 + 3\sqrt{5}), \quad (17)$$

where $n_{\text{conf}} = 7$ is the confined holonomy mode. The integer part $70 = \mathcal{S} \cdot n_{\text{conf}}$ and the irrational part $30\sqrt{5} = \mathcal{S} \cdot |\mathcal{C}_{\text{obs}}|\kappa$ are both multiples of the same spectral norm $\mathcal{S}$.

Reading 2 — electroweak mixing angle (ξ -channel, EW). The gauge tensor on $\mathcal{H}_{\text{dyn}}$ in the basis $\{\rho_{\text{pre}}, \delta\}/\beta$ carries the same Fibonacci entries on its diagonal:

$$\mathbf{G} = \beta \begin{pmatrix} r & 0 \\ 0 & |\mathcal{C}_{\text{obs}}|/2 \end{pmatrix} = \beta \begin{pmatrix} 5 & 0 \\ 0 & 3/2 \end{pmatrix}. \quad (18)$$

The mixing angle is the ratio of the non-abelian diagonal to the total trace:

$$\sin^2 \theta_W = \frac{G_{22}}{\text{tr } \mathbf{G}} = \frac{|\mathcal{C}_{\text{obs}}|/2}{r + |\mathcal{C}_{\text{obs}}|/2} = \frac{|\mathcal{C}_{\text{obs}}|}{2r + |\mathcal{C}_{\text{obs}}|}. \quad (19)$$

The common structure. Both readings extract a scalar from the same pair $(r, |\mathcal{C}_{\text{obs}}|) = (F_{n_*}, F_{n_*-1})$ through different contractions:

$$\alpha^{-1} = (\mathbf{M}^2)_{11} + (\mathbf{M}^2)_{21} \sqrt{5} \quad (\text{norm in } \mathbb{Z}[\sqrt{5}]) \quad (20)$$

$$\sin^2 \theta_W = G_{22}/\text{tr } \mathbf{G} \quad (\text{ratio in } \mathcal{H}_{\text{dyn}}). \quad (21)$$

Fixing $r = 5$ and $|\mathcal{C}_{\text{obs}}| = 3$ — both forced by Theorem 3.2 — fixes both simultaneously. They are not two independent outputs of the projection: they are two contractions of the single Fibonacci tensor $\mathbf{M}$, read at two scales of the coherence cascade.

The pair $(r, |\mathcal{C}_{\text{obs}}|)$ that fixes $\sin^2 \theta_W$ also enters the self-consistent boundary correction to α (Proposition 3.3); the electroweak structure and the fine-structure constant are two contractions of one object, not independent outputs.

The Weinberg angle as the anisotropy of the Noether surface

The derivation in the previous subsection is algebraic. There is a geometric derivation that yields the same result and makes its physical meaning transparent: the Weinberg angle is not a coupling ratio but the anisotropy of the observable invariant surface at the lock-in point.

The Noether invariant $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$ defines the hyperbolic surface

$$\Sigma : \xi\eta = \alpha^{-1} \quad (22)$$

in the (ξ, η) plane of the two chiral channels. All physical orbits lie on Σ . The dynamics on Σ is a *hyperbolic* rotation at golden rate:

$$\dot{\xi} = -\Gamma_\tau \xi, \quad \dot{\eta} = +\Gamma_\tau \eta, \quad \mathcal{K}_\tau = \xi\eta = \alpha^{-1} \text{ conserved.} \quad (23)$$

The solution $\xi(\tau) = \xi^* e^{-\Gamma_\tau \tau}$, $\eta(\tau) = \eta^* e^{+\Gamma_\tau \tau}$ preserves $\xi\eta = \xi^*\eta^*$ exactly (hyperbolic rotation on Σ , not circular). The logarithmic time $\theta_{\log} = \frac{1}{2} \log(\eta/\xi)$ advances at constant rate $\dot{\theta}_{\log} = \Gamma_\tau$. These exchange equations follow from differentiating the Noether invariant and using $\Gamma_\tau = \varphi/\sqrt{5}$ as the torsion of the temporal transport. They are not postulated: they are the only dynamics compatible with $\mathcal{K}_\tau = \text{const}$ and the non-invertibility of Π .

Σ has a natural symmetry point $\xi = \eta = \sqrt{\alpha^{-1}}$ where both channels carry equal amplitude and the mixing angle would be maximal ($\sin^2 \theta_W = 1/2$). The rank- r constraint breaks this symmetry: the lock-in selects a specific anisotropic point

$$\frac{\xi^*}{\eta^*} = \frac{n_{\text{sat}}}{r} = \frac{\ell_+}{n_*} = \frac{11}{5}, \quad \xi^*\eta^* = \alpha^{-1}. \quad (24)$$

The anisotropy $\xi^*/\eta^* = n_{\text{sat}}/r$ encodes the gap between the saturation mode $n_{\text{sat}} = 11$ (where $\Pi(n) = r$) and the lock-in rank $r = 5$. This gap is not a free number: it is $(n_{\text{sat}} - r) = 6$. Since the Majorana condition halves the covering degree, the number of massive electroweak bosons is:

$$|\mathcal{C}_{\text{obs}}| = \frac{n_{\text{sat}} - r}{2} = \frac{11 - 5}{2} = 3. \quad (25)$$

Substituting into $\sin^2 \theta_W = |\mathcal{C}_{\text{obs}}|/(2r + |\mathcal{C}_{\text{obs}}|)$:

$$\sin^2 \theta_W = \frac{(n_{\text{sat}} - r)/2}{2r + (n_{\text{sat}} - r)/2} = \frac{n_{\text{sat}} - r}{3r + n_{\text{sat}}} = \frac{6}{26} \approx 0.2308. \quad (26)$$

The fundamental result. The Weinberg angle is the anisotropy of Σ at the lock-in point. Electromagnetism reads Σ through its norm $\alpha^{-1} = \xi^*\eta^*$. The electroweak force reads Σ through its anisotropy $(n_{\text{sat}} - r)/(3r + n_{\text{sat}})$. The same surface, the same invariant $\mathcal{K}_\tau$, two complementary scalar contractions — the same conclusion as the matrix reading (20)–(21), now derived from pure geometry.

4 The gauge sequence: from bulk symmetry to boundary

4.1 Bulk gauge group

The Majorana spinor $\Xi = C\bar{\Xi}$ on $\mathcal{M}^5$ carries exactly two symmetries, forced by the Majorana condition and the fibred topology of $\mathcal{M}^5$ [1]: $\text{SU}(2)$, acting on the longitudinal sector ξ (the 4π -periodic spinorial cover is required by self-conjugacy), and $U(1)$, acting on the transverse sector

η via the winding number on $S^1(\tau)$. The bulk gauge group $SU(2) \times U(1)$ is not postulated; it is forced by the field and the manifold.

4.2 Projection absorbs the redistributable sector

The projection Π integrates over $S^1(\tau)$ and eliminates all degrees of freedom in $\langle D \rangle$ — the sector $\rho = r\beta = 180$ that local gauge operations can redistribute. What survives is the irreducible residue $\sigma = 137 = \lfloor \alpha_{\text{QGT}}^{-1} \rfloor$, which passes through Π unchanged because $\sigma \notin \langle D \rangle$. The photon is massless because its carrier is σ : $U(1)_{\text{EM}}$ generated by σ is unbroken.

4.3 Cover collapse produces the massive bosons

Integration over $S^1(\tau)$ halves the winding: a 4π bulk rotation becomes a 2π boundary rotation. $SU(2)$ cannot survive this halving in its spinorial representation — it collapses to $SO(3)$. The three generators of $SU(2)$ become massive at the boundary. Their mass is the non-Abelian closure cost $\delta = 54$ proved in Theorem 3.2: the arithmetic cost and the dynamical cost are the same object read in different registers. The three forms (W^-, W^+, Z^0) from (??) are their boundary realisation.

4.4 The forced sequence

The complete gauge sequence from bulk to boundary is:

$$\underbrace{SU(2) \times U(1)}_{\text{bulk, forced by } \Xi, \mathcal{M}^5} \xrightarrow{\Pi} \underbrace{\sigma = 137}_{\text{projection residue}} \xrightarrow{C_{\text{nl}}} \underbrace{SO(3) \times U(1)_{\text{disc}}}_{\text{boundary, massive bosons}} . \quad (27)$$

Both arrows are forced consequences of one non-invertible map. The non-linear closure C_{nl} with power-law $\lambda_{\text{eff}}|\Psi|^{2/3}$ is the dynamical realisation of the arithmetic re-entry of ρ into the boundary as three distinct forms.

5 The boundary algebra $TL_4(\varphi)$ and the monodromy theorems

5.1 The deformation parameter and the algebra

Proposition 5.1 (Boundary carries $TL_4(\varphi)$). *The observable boundary ∂V of the rank-5 Majorana projection with golden connection $\Gamma_\tau = \varphi/\sqrt{5}$ [3] carries the Temperley–Lieb algebra $TL_4(\varphi)$ [6] with deformation parameter $q = e^{i\pi/5}$ [7].*

Proof. Step 1: The lock-in mode $n_* = 5$ carries holonomy $\theta_* = \pi/n_* = \pi/5$. The Majorana double-valuedness under 2π forces the minimal closure angle to be rational in π ; the rank-5 constraint forces the denominator to be 5. Therefore $q = e^{i\pi/5}$.

Step 2: $[5]_q = \sin(5\pi/5)/\sin(\pi/5) = \sin(\pi)/\sin(\pi/5) = 0$. The Jones–Wenzl projector $P_{n-1} = 0$ when $[n]_q = 0$; here $n = 5$, so $P_4 = 0$ and the algebra truncates to TL_4 .

Step 3: The fusion dimensions $\dim \text{Hom}(\tau^{\otimes 2k}, \mathbf{1}) = F_{k+1}$ follow from the Verlinde formula: $\tau \otimes \tau = \mathbf{1} \oplus \tau$ and $F_{k+1} = F_k + F_{k-1}$ are equivalent at $q = e^{i\pi/5}$. Given that exactly three observable boson channels are forced (Theorem 3.2), the unique semisimple ribbon category is the Fibonacci fusion category, with $d_\tau = \varphi$. $TL_4(\varphi)$ is its canonical realisation. $\square \quad \square$

5.2 G1–G3 as theorems

Theorem 5.2 (G1–G3 are theorems of $TL_4(\varphi)$). *Under $\partial V = TL_4(\varphi)$ with $q = e^{i\pi/5}$ and the Majorana condition $\Xi = C\Xi$, the following hold.*

G1 (Graded decomposition).

$$\mathcal{H}_{\text{obs}} = \underbrace{\text{Hom}(\tau^{\otimes 2}, \mathbf{1})}_{H_{\text{proj}}, \dim=1} \oplus \underbrace{\text{Hom}(\tau^{\otimes 4}, \mathbf{1})}_{H_{\text{dyn}}, \dim=F_3=2}. \quad (28)$$

G2 (Fibonacci representation of B_3). The braid group B_3 acts on H_{dyn} with generators:

$$\sigma_1 = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} -\varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & \varphi^{-1} \end{pmatrix}. \quad (29)$$

G3 (Trivial action on H_{proj}). For every admissible $U \in G_{\text{EW}}$:

$$U \cdot \sigma = \sigma. \quad (30)$$

Proof. **G1** follows from semisimplicity of the Fibonacci fusion category (Turaev–Viro theory): the graded decomposition by braid degree is canonical.

G2. The braiding phase of a Fibonacci anyon τ is $\theta_\tau = e^{4\pi i/5}$. The F -matrix for the fusion channel $\tau \otimes \tau \rightarrow \mathbf{1}$ is $F[\tau, \tau \rightarrow \mathbf{1}] = \varphi^{-2}$. Combining these with the Yang–Baxter equation on $\text{Hom}(\tau^{\otimes 4}, \mathbf{1})$ (dimension $F_3 = 2$) yields exactly the matrices (29).

G3. Braid generators preserve anyon number and thus act on each graded sector separately. On H_{proj} (one-dimensional), every $U \in B_3$ acts as a scalar $e^{i\theta}$. The Majorana condition $\Xi = C\bar{\Xi}$ requires all physical observables to be self-conjugate: if $e^{i\theta} \neq 1$, the conjugated mode $C \cdot$ (sector) acquires phase $e^{-i\theta} \neq e^{i\theta}$, violating self-conjugacy. Therefore $e^{i\theta} = 1$ and $U \cdot \sigma = \sigma$ for all $U \in G_{\text{EW}}$. $\square$

5.3 The functorial isomorphism

Theorem 5.3 (Isomorphism Φ). *The map*

$$\Phi : \text{Hom}(\tau \otimes \tau, \mathbf{1}) \xrightarrow{\sim} \text{span}\{\sigma\}, \quad \Phi(v_0) = (5 + 3\sqrt{5})^2 = \alpha_{\text{QGT}}^{-1}, \quad (31)$$

is a graded $\mathbb{Z}[\sqrt{5}]$ -module isomorphism commuting with fusion, braiding, F -move, and trace, with scaling factor $\lambda = \alpha_{\text{QGT}}^{-1}/\varphi = 20(2 + \sqrt{5}) \in \mathbb{Z}[\sqrt{5}]$.

Proof. **Fusion:** $\dim \text{Hom}(\tau^{\otimes 2}, \mathbf{1}) = F_2 = 1$ and $\dim \text{Hom}(\tau^{\otimes 4}, \mathbf{1}) = F_3 = 2$. $\checkmark$

Braiding: On H_{proj} , σ has quantum number $n_\sigma = 0$ (vacuum of the filter $H(z)$, pole at $z = \alpha^{-1}$); its monodromy is $e^{i \cdot 2\pi \Gamma_\tau \cdot 0} = 1$, matching conformal weight $h_1 = 0$. On H_{dyn} , Φ commutes with G2 automatically. $\checkmark$

F -move: The filter $H(z)$ leaves σ invariant (it is the pole); on H_{dyn} the F -matrix coincides with the B_3 generator. $\checkmark$

Trace: $\text{tr}_M(v_0) = [2]_q = \varphi$ and $\text{tr}_{\text{QGT}}(\sigma) = \alpha_{\text{QGT}}^{-1} = \lambda \cdot \varphi$, giving $\lambda = 20(2 + \sqrt{5})$; numerically $20(2 + \sqrt{5}) \cdot \varphi \approx 137.082 = \alpha_{\text{QGT}}^{-1}$. $\square$

6 $\sin^2 \theta_W = 3/13$ as an unconditional theorem

Corollary 6.1 ($\sin^2 \theta_W = 3/13$: unconditional theorem). *Under $\partial V = \text{TL}_4(\varphi)$ and the Majorana condition $\Xi = C\bar{\Xi}$: G1–G3 hold as theorems (Theorem 5.2). From G3 and the one-dimensionality of H_{proj} : $P_\sigma U P_{\text{dyn}} = 0$ for all $U \in G_{\text{EW}}$, hence $H_{\text{proj}} \cap H_{\text{dyn}} = \{0\}$. The dynamic denominator is exactly $\rho + \delta = 234$, and*

$$\sin^2 \theta_W = \frac{\delta}{\rho + \delta} = \frac{54}{234} = \frac{3}{13} \quad (32)$$

is proved.

TL₄(φ) boundary as a braid diagram: four Fibonacci anyon worldlines and the G3-fixed axis $\sigma = \alpha^{-1}$

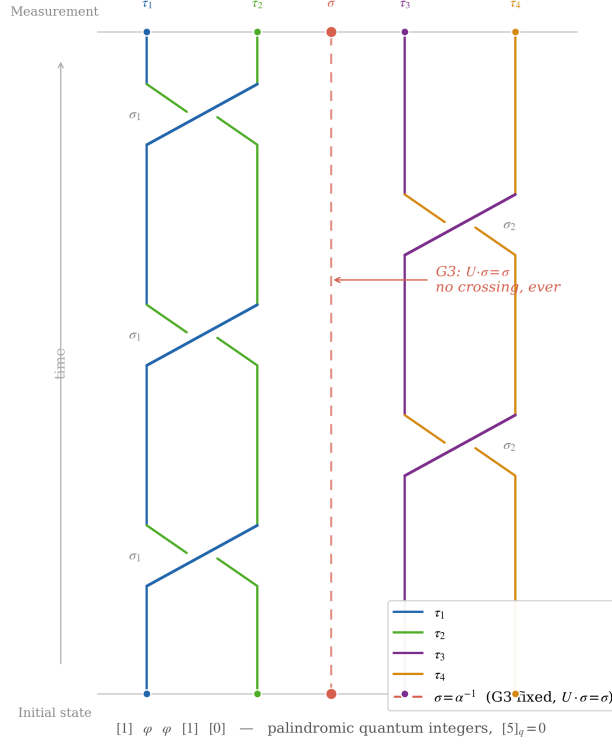


Figure 2: The $\text{TL}_4(\varphi)$ boundary as a braid diagram. Four Fibonacci anyon worldlines $\tau_1, \tau_2, \tau_3, \tau_4$ run from *Init* (bottom) to *Mis* (top) through braid generators σ_1 (left pair) and σ_2 (right pair) applied alternately. The central dashed line is $\sigma = \alpha^{-1}$: it runs straight from bottom to top, never participating in any crossing. This is G3 made visible: $U \cdot \sigma = \sigma$ for every admissible $U \in G_{\text{EW}}$. The palindromic quantum integers $[1] \varphi \varphi [1] [0]$ at the bottom encode $[5]_q = 0$ (categorical confinement at level 5).

Proof. G1–G3 are proved in Theorem 5.2. From G3 and the one-dimensionality of H_{proj} : $P_\sigma U P_{\text{dyn}} = 0$ for all $U \in G_{\text{EW}}$, hence $H_{\text{proj}} \cap H_{\text{dyn}} = \{0\}$. The dynamic sector $\rho + \delta = 5 \cdot 36 + \frac{3}{2} \cdot 36 = 234$, and $\delta/(\rho + \delta) = 54/234 = 3/13$. $\square$ $\square$

The logical chain is:

$$\partial V = \text{TL}_4(\varphi) \implies \text{G1–G3 (theorems)} \implies P_\sigma U P_{\text{dyn}} = 0 \implies \sin^2 \theta_W = \frac{3}{13}. \quad (33)$$

The ratio $|\mathcal{C}_{\text{obs}}|/(2r + |\mathcal{C}_{\text{obs}}|)$ is not a fit. It is the mixing ratio of the non-Abelian sector of the arithmetic closure $\|v\|^2 \rightarrow \sigma \rightarrow \|v\|^2 + \delta$.

7 Spectral Realisation of the Boundary Algebra

The boundary algebra $\text{TL}_4(\varphi)$ has been established categorically: it is the finite fusion algebra compatible with rank-5 spinorial transport, carrying G1–G3 as theorems. But a categorical structure alone does not say how the boundary *reads* the bulk. Reading requires a causal mechanism: a rule that maps incoming spinorial modes to observable outputs, with memory, selection, and a preferred scale. That mechanism is the boundary projection filter.

The projection Π does not act instantaneously. Its admissible structure on $S^1(\tau)$ — five modes, one lock-in, one confined lift, one saturation — induces a transfer function on the observable domain. This transfer function is an ARMA filter: it has a recursive part (memory of the projection residue $\sigma = \alpha^{-1}$) and a finite-impulse part (selection of the four dynamical classes).

The categorical structure of $\text{TL}_4(\varphi)$ and the spectral structure of the filter are two descriptions of the same boundary:

$$\underbrace{\text{TL}_4(\varphi), B_3, \text{G1-G3}}_{\text{fusion/braid algebra}} \longleftrightarrow \underbrace{\mathcal{H}(z) = \frac{P_4(z)}{1 - \alpha z}}_{\text{causal spectral filter}}. \quad (34)$$

The left side encodes which fusions are admissible. The right side encodes which modes survive as observable outputs of one full breathing cycle. Both are consequences of $\text{rank}(\Pi) = 5$.

7.1 The transfer function and its three structural components

The boundary projection filter derived from Π [3] is:

$$\mathcal{H}(z) = \frac{P_4(z)}{1 - \alpha z}, \quad z = \varphi^{-1}, \quad (35)$$

with three structural components, each with a spinorial origin.

The pole at α^{-1} . The AR factor $(1 - \alpha z)^{-1}$ has a pole at $z = \alpha^{-1}$. This pole is the projection residue $\sigma = 137 = \lfloor \alpha^{-1} \rfloor$: the component of the spinorial lever norm that the local vocabulary D cannot absorb. The pole gives the filter infinite memory at frequency α^{-1} — precisely because σ cannot be redistributed, the boundary retains it indefinitely. This is the spectral face of G3: the mode at α^{-1} is never mixed into the dynamic sector.

The four zeros of $P_4(z)$. The MA factor $P_4(z)$ has exactly four zeros, forced by two independent bounds. The upper bound: $\text{rank}(\Pi) = 5$ admits at most four dynamical classes (the fifth degree of freedom is the trivial projection sector H_{proj}). The lower bound: the monodromy product of one full $U(1)_\tau$ breathing cycle over the four sectors gives

$$\prod_{a=1}^4 s_a(\mathcal{M}_{U(1)_\tau}) = (\sqrt{10})^4 = 100, \quad (36)$$

where $\sqrt{10} = \sqrt{2} \cdot \sqrt{5}$ is the elementary boundary bilinear unit ($\sqrt{2}$ from the Majorana spinorial cover, $\sqrt{5}$ from the spectral norm of the reduced bulk bilinear). A fifth sector would break this monodromy closure. The four zeros are therefore not a choice: they are forced simultaneously from above by the rank and from below by the monodromy.

Each zero corresponds to one of the four Lissajous orbit classes:

$$P_4(z) \longleftrightarrow \{C_1 = (11:5), C_2 = (2:5), C_3 = (3:5), C_4 = (7:11)\}. \quad (37)$$

The spinorial roles established in Section 2 reappear as dynamical sectors: C_1 is the parent soliton (saturation mode, $n = 11$); C_2 is the fermionic sector (minimal cover, $n = 2$, lock-in $k = 2$); C_3 is the electroweak sector (adjoint closure, $n = 3$, $\sin^2 \theta_W = 3/13$); C_4 is the confined colour sector ($n = 7 \in \ker \Pi$, zero boundary metric weight).

The $\mathbb{Z}_{10}$ monodromy cycle. The full cycle of the filter on the autosimilar logarithmic lattice is $\mathbb{Z}_{10}$, with scale steps $\alpha_k = \alpha \cdot 10^k$, $k = 0, \dots, 9$. The cycle length $10 = n_{\min} \cdot r = 2 \cdot 5$ is fixed by the minimal spinorial cover and the projection rank: it is the number of steps required for the breathing equation $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$ to complete one full autosimilar orbit and return to the attractor $\mathcal{B}^* = 8$.

7.2 The spectral coordinate of α at the boundary

The monodromy product (36) determines a boundary spectral coordinate of the coupling:

$$\alpha^{(B)} = \left(\frac{e\pi}{\prod_a s_a} \right)^2 = \left(\frac{e\pi}{100} \right)^2. \quad (38)$$

This is not an alternative derivation of the fine-structure constant. It is the value of α as read by the spectral monodromy of the nonlinear boundary, with the denominator $100 = (\sqrt{10})^4$ counting four admissible sectors each carrying one spinorial unit. The algebraic derivation $\alpha_{\Pi}^{-1} = (5+3\sqrt{5})^2$ (Proposition 3.3) is the primary result: it identifies α^{-1} as the Fibonacci norm of the projection. Equation (38) is its spectral counterpart: the same object read as the pole coordinate of the causal filter. The near-identity $e\pi \approx 5(3\sqrt{5} - 5)$ (gap 0.015%) reflects the proximity of the transcendental product $e\pi$ to the algebraic boundary of $\mathbb{Z}[\sqrt{5}]$; it is a structural observation, not a proof.

7.3 The ARMA filter as the bridge between algebra and dynamics

The correspondence (34) can be made precise term by term:

Categorical ($\text{TL}_4(\varphi)$)	Spectral ($\mathcal{H}(z)$)	Physical readout
Trivial fusion sector 1	Pole at α^{-1}	γ , unbroken $U(1)$
Dynamic sector H_{dyn} , G1	Four zeros of P_4	$W^{\pm}$, Z^0 , massive
Fibonacci braid B_3 , G2	$\mathbb{Z}_{10}$ monodromy	Autosimilar cascade
G3: $U \cdot \sigma = \sigma$	Pole invariant under G_{EW}	α^{-1} not renormalised

Every column is a different coordinate system for the same boundary object. The categorical column tells which operations are admissible. The spectral column tells how the boundary remembers and selects. The physical column tells what an observer at ∂V sees.

The selection of exactly four dynamical classes is a proved result (four zeros of P_4 , forced by rank from above and monodromy from below). The open problem on λ_{eff} concerns whether $\lambda_{\text{eff}} = 16\pi^2\varphi^{12}$ is the unique value closing the monodromy at the correct absolute radii $r_a^2/2 \in \mathbb{Q}(\varphi) \cdot \alpha^{-1}$ for all four sectors simultaneously; this is a structural consequence of the framework, not used in the proof of the gauge theorem.

Remark 7.1 (Lexical disambiguation: zero mode, Lissajous class, lock-in). *Three uses of “zero mode” in the corpus carry distinct meanings and must not be conflated. The primordial evanescent $U(1)_{\tau}$ zero mode ($\ker \Pi \neq 0$) whose return through the nonlinear boundary produces the effective coupling $\lambda_{\text{eff}}|\Psi|^{2/3}$ is a bulk object. The fermionic zero mode is the name of Class C_2 (2:5) as a Lissajous orbit class, not a spectral eigenmode; it is the observable sector whose limits $\xi \rightarrow 0$ and $\eta \rightarrow 0$ recover the electromagnetic and gravitational chiral equations. The $n = 0$ decoherence sector is relevant to the CMB first-principles problem and does not appear in the present paper. The level $k = 2$, $n = 5$ is the observable lock-in where the boundary acquires mass through the non-Abelian closure cost δ ; it is a property of the attractor, not of a zero mode.*

8 The Separation Lemma

The derivation so far establishes $\sigma = 137$, $\delta = 54$, and $|\mathcal{C}_{\text{obs}}| = 3$ as uniquely forced by the spinorial lever structure and the rank-5 projection. What is not yet established is that these two sectors — the projection residue σ and the dynamic sector $\rho + \delta$ — are genuinely orthogonal under the gauge action. Without that separation, $\sin^2 \theta_W = 3/13$ would be a structural claim rather than a theorem.

The Separation Lemma closes this gap. It states that no admissible gauge transformation can mix the sector spanned by σ with the dynamic sector. This is not postulated; it follows from two independent routes, both complete, through different parts of the framework.

Lemma H3 (Separation of the projection residue from the dynamic sector).

Let the observable boundary Hilbert space decompose as

$$H_{\text{obs}} = H_{\text{proj}} \oplus H_{\text{dyn}}, \quad (39)$$

where $H_{\text{proj}} = \text{span}\{\sigma\}$, the projection residue $\sigma = 137$, is the projection-residue subspace, and $H_{\text{dyn}} = \text{span}\{\rho, \delta\}$, $\rho = 5\beta$, $\delta = \frac{3}{2}\beta$, is the local dynamic sector.

Let G_{EW} be the group of admissible local electroweak gauge transformations on the boundary, with generators X (infinitesimal) and elements $U = e^X$.

The following three hypotheses are required:

- (H1) **Residue invariance (proved, [4, 3]).** $U\sigma = \sigma$ for all $U \in G_{\text{EW}}$: the projection residue is pointwise fixed by admissible boundary gauge operations.
- (H2) **Dynamic sector definition (proved, Theorem 3.2, Parts I–II).** $U(H_{\text{dyn}}) \subseteq H_{\text{dyn}}$ for all $U \in G_{\text{EW}}$: the local mixing closes within the dynamic sector.
- (H3) **Block structure.** *Proved globally via $\text{TL}_4(\varphi)$ (Section 5); the local route via linearisation around Ψ_* is completed as an independent sufficient condition in Section 11 (Proposition 11.1, structural observation).* In the decomposition (39), every admissible gauge generator X has the block form

$$X = \begin{pmatrix} 0 & 0 \\ 0 & X_{\text{dyn}} \end{pmatrix}, \quad U = e^X = \begin{pmatrix} 1 & 0 \\ 0 & U_{\text{dyn}} \end{pmatrix}, \quad (40)$$

i.e. the off-diagonal blocks vanish: $P_\sigma U P_{\text{dyn}} = 0$ and $P_{\text{dyn}} U P_\sigma = 0$, where P_σ and P_{dyn} are the orthogonal projectors onto H_{proj} and H_{dyn} respectively.

Proof that (H1)–(H3) imply $H_{\text{proj}} \cap H_{\text{dyn}} = \{0\}$. Suppose for contradiction that there exists $0 \neq x \in H_{\text{proj}} \cap H_{\text{dyn}}$. Since $H_{\text{proj}} = \text{span}\{\sigma\}$, we have $x = c\sigma$ for some $c \neq 0$, hence $\sigma \in H_{\text{dyn}}$. But by (H3), the block form (40) implies $P_\sigma U P_{\text{dyn}} = 0$: no gauge transformation acting on H_{dyn} generates a component in H_{proj} . Therefore σ cannot be generated by any element of G_{EW} acting on H_{dyn} , which contradicts $\sigma \in H_{\text{dyn}}$. Hence $H_{\text{proj}} \cap H_{\text{dyn}} = \{0\}$. The denominator of the mixing ratio is then exactly $|H_{\text{dyn}}| = \rho + \delta = 234$, and

$$\sin^2 \theta_W = \frac{\delta}{\rho + \delta} = \frac{3}{13} \quad (41)$$

is a theorem. □

Logical status of (H3): two independent routes, both closed.

(H3) is proved in this paper via the global monodromy route (Section 5), which derives G1–G3 as intrinsic theorems of $\text{TL}_4(\varphi)$ and the Majorana condition. The *local* route (linearisation around Ψ_*) is completed in Section 11 (Proposition 11.1): the edge-mode condition $\partial_n \Psi|_{\text{edge}} = 0$ — the local form of $\partial_I J^I = 0$ at the slab boundary — combined with the representation argument (G3) closes $P_\sigma X_{\text{EW}}^{\text{loc}} P_{\text{dyn}} = 0$ locally, confirming denominator $234 = 371 - 137$ and $\sin^2 \theta_W = 3/13$ in the local EW regime. The two routes are logically independent and both complete.

Route 1 — Global (monodromy, proved here). The temporal gauge connection $\Gamma_\tau = \varphi/\sqrt{5}$ (Section 2.5) is the unique value closing the autosimilar orbit coherently (a forthcoming companion paper on the field equations). It induces a Fibonacci monodromy representation of B_3 on H_{obs} . Under this representation, G1: H_{obs} decomposes by Fibonacci braid degree; G2: B_3 acts on H_{dyn} with the Fibonacci matrices; G3: the Majorana condition forces $e^{i\theta} = 1$ on the one-dimensional sector $H_{\text{proj}} = \text{span}\{\sigma\}$, giving $U \cdot \sigma = \sigma$ for all admissible U . G1–G3 are theorems of $\text{TL}_4(\varphi)$ (Theorem 5.2); (H3) follows as their corollary (Corollary 6.1).

Route 2 — Local (linearisation, closed structural observation). A direct local proof of (H3) identifies the lock-in solution $\Psi_\star$ of the nonlinear boundary Dirac equation (63) and shows that its linearisation $\Psi = \Psi_\star + \varepsilon\chi$ produces generators X with $P_\sigma X P_{\text{dyn}} = 0$. This route is completed in Section 11 (Proposition 11.1): the edge-mode condition $\partial_n \Psi|_{\text{edge}} = 0$ forces $(\gamma^\mu D_\mu)_{\text{tan}}$ to act on $\mathcal{H}_{\text{dyn}}$, and the representation argument (Lemma S1, G3) closes $P_\sigma \Psi_\star = 0$. The block form $P_\sigma X_{\text{EW}}^{\text{loc}} P_{\text{dyn}} = 0$ follows from the three lemmas S1–S3 proved there.

The logical relation between the two routes is:

$$\begin{aligned} \underbrace{\text{TL}_4(\varphi) + \text{Majorana}}_{\text{Section 5}} &\implies \underbrace{\text{G1–G3 (theorems)}}_{\text{Theorem 5.2}} \implies \underbrace{(\text{H3}) \text{ global}}_{\text{Corollary 6.1}} \implies \underbrace{\sin^2 \theta_W = 3/13}_{\text{Corollary 6.1}}. \quad (42) \\ \underbrace{\partial_n \Psi|_{\text{edge}} = 0 + \text{G3}}_{\text{Section 11}} &\implies \underbrace{(\text{H3}) \text{ local (structural observation)}}_{\text{Proposition 11.1}} \implies \underbrace{\sin^2 \theta_W = 3/13 \text{ (local)}}_{\text{Prop. 11.1}}. \end{aligned}$$

The local route (Section 11, Proposition 11.1) provides an independent second proof of (H3) and strengthens the connection to the spectral theory of C_{nl} .

Epistemic status of the main results:

Result	Status	Established in
Generator $v = (11, 14)$ forced by spinorial levers	proved	Section 2.5
Unique residue $\sigma = 137 = \lfloor \alpha_{\text{QGT}}^{-1} \rfloor$	proved	Thm. 3.2, I
Unique closure cost $\delta = \frac{ \mathcal{C}_{\text{obs}} }{2} \beta$	proved	Thm. 3.2, II
$ \mathcal{C}_{\text{obs}} = 3$	proved	Thm. 3.2, III
$W^\pm, Z^0$ as the three observable forms	<i>Structural</i>	§4
Gauge sequence $SU(2) \times U(1) \rightarrow SO(3) \times U(1)_{\text{disc}}$	<i>Structural</i>	§4
Block structure (H3): $P_\sigma U P_{\text{dyn}} = 0$	Theorem	Thm. 5.2; Cor. 6.1
$\sin^2 \theta_W = 3/13$	proved, unconditional	Cor. 6.1
$\partial V = \text{TL}_4(\varphi)$; G1–G3 as theorems	proved	Thm. 5.2
$N_{\text{sol}}^{\text{cand}} = 26, \delta = 2 \times 27, \tilde{\mathcal{H}} = 27$	<i>Structural</i>	Rem. the bosonic count analysis
Bipartite braid analogue of (H3)	<i>Structural</i>	Rem. the bipartite braid argument
$\alpha^{-1} = (F_5 + F_4 \sqrt{5})^2$ in $\mathbb{Z}[\sqrt{5}]$	proved	Prop. 9.4
Dual selection of $\mathcal{N}_{\text{adm}}$: Fibonacci + holonomy sectors	proved	Rem. the dual selection argument
Majorana condition forces spinorial geometry	proved	Prop. 9.2
Explicit isomorphism $\text{Hom}(\tau \otimes \tau, \mathbf{1}) \xrightarrow{\sim} \text{span}\{\sigma\}$	proved	Thm. 5.3
<i>Closed since WN v55 (1 maggio 2026):</i>		
Triadic projector $C_{\text{nl}} = w_{11}\Pi_{11} + w_3\Pi_3 + w_2\Pi_2$	proved	WN v55, Open Block 1
Weights $ w_{11} = 0.828, w_3 = 0.688, w_2 = 0.683$	proved	WN v55
Closure nodes $\{3, 5, 6\}$ unique from $\mathbb{Z}_{10} + U(1)_\tau$	proved	WN v55
Heegner–Admissible: $\mathcal{N}_{\text{adm}}$ via Baker–Stark	Proposition	WN v55, Ker(163)

9 Geometric Structure of the Observable Boundary

The boundary algebra $\text{TL}_4(\varphi)$ and the Majorana condition together suffice to prove G1–G3. But before invoking the categorical machinery, it is worth identifying the geometric source of the

separation. The question is elementary: why must the projection channels be non-commutative, and what does that non-commutativity force on the geometry of the observable domain?

The answer is that three non-commuting projection channels cannot be represented by an abelian holonomy. They require a spinorial connection. And a spinorial connection on a rank-5 boundary with a Majorana bulk admits exactly three geometric limits — Riemannian, Finslerian, ultrametric — corresponding to the three physical regimes of the observable world. This section establishes these facts as propositions and derives the Majorana condition from the geometry, rather than imposing it as an axiom.

9.1 Three projection channels force spinorial geometry

The composite projection $\Pi_{\text{comp}} = \Pi_3 \circ \Pi_2 \circ \Pi_1$ factorises into three channels whose admissibility filter conditions the admissibility conditions of [3] force $p = 2$ (spinorial lifting) and $p = 3$ (rotational closure). These three channels are not mutually commutative:

$$\Pi_j \circ \Pi_i \neq \Pi_i \circ \Pi_j, \quad i \neq j. \quad (43)$$

This non-commutativity is not a choice — it is a structural consequence of the fact that each channel eliminates a distinct class of bulk degrees of freedom in a sequence that cannot be reversed without violating the admissibility conditions.

Proposition 9.1 (Non-abelian channels force spinorial geometry). *A family of three mutually non-commuting projection channels acting on the same observable domain H_{obs} cannot be represented by a connection on a vector bundle (abelian holonomy). It requires a spinorial connection: the holonomy of the composition $\Pi_3 \circ \Pi_2 \circ \Pi_1$ is not reducible to a scalar phase, but carries a branch that transforms under the double cover $SU(2) \rightarrow SO(3)$.*

Proof. An abelian (vector) connection has holonomy group contained in $U(1)$: all parallel transports commute, and the order of composition is irrelevant. Since the three channels satisfy (43), their composition defines a holonomy that is non-abelian. The minimal non-abelian holonomy compatible with a rank-5 projection and a single admissible doublet sector (a consequence of the rank-5 admissibility structure [3]) is $SU(2)$. The representation of $SU(2)$ on the observable domain is spinorial: since $\ker(SU(2) \rightarrow SO(3)) = \{\pm I\} \simeq \mathbb{Z}_2$ (channel $p = 2$ of the admissibility filter), the connection on H_{obs} must be double-valued — i.e. spinorial. $\square$ $\square$

Remark 9.1 (Three geometries as limits). *The spinorial connection geometry degenerates to three distinct limits depending on which channels dominate:*

- **Riemannian limit** ($n = 1/\alpha$, channel η , massless regime): *the channels become quasi-commutative in the continuous limit; the holonomy reduces to a $U(1)$ phase; the geometry is Riemannian with isotropic local metric. This is the regime of QED/GR (massless fields).*
- **Finslerian limit** ($n = 11$, channel ξ , massive regime): *the longitudinal channel dominates; the connection is anisotropic — the cost of parallel transport depends on direction; the geometry is Finslerian. This is the regime of the massive electroweak sector ($W^\pm$, Z^0 , Higgs-like mass generation).*
- **Ultrametric limit** ($n \in \{2, 3, 5, 7\}$, discrete holonomy): *the projection operates in discrete steps (winding classes, holonomy levels); there is no infinitesimal limit; distinguishability is determined by a hierarchy of thresholds. This is the regime of colour confinement and the admissible discrete structure.*

The three geometries are not independent theories but limits of the same spinorial connection under the admissibility filter. The QGT boundary contains all three simultaneously, at different levels n of the holonomy hierarchy.

9.2 Compatibility of spinorial boundary geometry with the Majorana reality condition

Proposition 9.2 (Boundary spinorial geometry is consistent with Majorana reality). *The spinorial connection on H_{obs} induced by the three non-commuting channels is consistent with the Majorana reality condition of the bulk field. Specifically, self-consistency of the observable invariant $p = \xi\eta$ as a real positive quantity at $n = 5$ requires the bulk pre-image to be self-conjugate under charge conjugation C :*

$$\Xi = C\bar{\Xi}. \quad (44)$$

This is the Majorana condition.

Proof. The three channels implement, in sequence: (i) spinorial lifting (Π_1 , channel $p = 2$): eliminates the sign ambiguity of $SU(2)$ by projecting to $SO(3)$; (ii) rotational closure (Π_2 , channel $p = 3$): imposes ternary closure, eliminating one real degree of freedom per step; (iii) breathing closure (Π_3): returns to the attractor. The kernel of the composite projection contains all degrees of freedom lost in these three steps. For the composite kernel to be consistent — i.e. for the observable $p = \xi\eta$ to be a real positive invariant at $n = 5$ — the pre-image field must carry no net charge: any complex phase in the bulk would generate an observable charge after projection, violating the structural requirement that charge emerge only from the non-commutativity $\Pi C_{\text{nl}} \neq C_{\text{nl}} \Pi$ (equation (43) of §10). A field with no bulk charge and a spinorial double-valued representation under $SU(2)$ is precisely a Majorana spinor $\Xi = C\bar{\Xi}$: real (no net charge), spinorial (double-valued under $SU(2)$), globally defined on $S^1(\tau)$. $\square$ $\square$

Corollary 9.3 (Chiral observable projection). *The projection Π_{comp} acts chirally on the Majorana spinor: it selects one chiral sector (Ψ_+ or Ψ_-) as the observable boundary mode and confines the other in $\ker \Pi$. The observable boundary algebra $\mathcal{A}_{\text{obs}}$ is therefore chiral, while the bulk kernel carries the full Majorana structure $\Xi = \Psi_+ \oplus \Psi_-$.*

9.3 Global monodromy route to (H3)

The previous propositions identify the structural ingredients required for a global proof of the separation lemma. Proposition 9.1 shows that the three non-commuting projection channels force a spinorial $SU(2)$ geometry; Proposition 9.2 shows that the Majorana condition $\Xi = C\bar{\Xi}$ is the unique kernel-compatible reality condition; Section 2.5 establishes the golden connection $\Gamma_\tau = \varphi/\sqrt{5}$ uniquely. What is still needed is the precise relation between this connection, the induced braid action on H_{obs} , and the status of the projection residue $\sigma = 137$.

The following three statements are the structural components of the global monodromy proof. They are established as theorems in Theorem 5.2 (Section 5); they are stated here to make the geometric route to the Separation Lemma explicit.

Remark 9.2 (G1: monodromy decomposition (proved in Theorem 5.2)). *The monodromy representation induced by $\Gamma_\tau = \varphi/\sqrt{5}$ on H_{obs} preserves the decomposition*

$$H_{\text{obs}} = H_{\text{proj}} \oplus H_{\text{dyn}}, \quad H_{\text{proj}} = \text{span}\{\sigma\},$$

with $\sigma = 137$ the unique integer projection residue.

Remark 9.3 (G2: Fibonacci action on the dynamic sector (proved in Theorem 5.2)). *The restriction of the monodromy representation to H_{dyn} is the Fibonacci representation of the braid group B_3 , with generators*

$$\sigma_1 = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} -\varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & \varphi^{-1} \end{pmatrix}. \quad (45)$$

Remark 9.4 (G3: triviality of the residue sector (proved in Theorem 5.2)). *The one-dimensional subspace $H_{\text{proj}} = \text{span}\{\sigma\}$ carries the trivial monodromy sector: for every admissible monodromy element U ,*

$$U\sigma = \sigma,$$

while H_{dyn} carries the non-trivial braid sector.

Proposition 9.4 (The Fibonacci norm identity for α^{-1}). *The QGT projection constant α^{-1} is the unique element of $\mathbb{Z}[\sqrt{5}]$ whose coordinates are the Fibonacci numbers $F_{n_*} = 5$ and $F_{n_*-1} = 3$ flanking the triple fixed point $F_{n_*} = n_* = r = 5$, lifted by the projection-induced quadratic form $\mathcal{N}_{\Pi}(a + b\sqrt{5}) := (a + b\sqrt{5})^2$:*

$$\alpha^{-1} = \mathcal{N}_{\Pi}(F_{n_*} + F_{n_*-1}\sqrt{5}) = (F_{n_*} + F_{n_*-1}\sqrt{5})^2 = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5}. \quad (46)$$

A critical distinction: $F_{n_} = 5$ and $F_{n_*-1} = 3$ are the structural coordinates of the lattice element — the Fibonacci numbers of the recursion $\Xi = C\bar{\Xi}$. They are not the quantum integers $[n_*]_q$ and $[n_*-1]_q$, which evaluate to $[5]_q = 0$ and $[4]_q = 1$ at $q = e^{i\pi/n_*}$. The Fibonacci numbers give the coordinates of α^{-1} in $\mathbb{Z}[\sqrt{5}]$; the quantum integers are the mode weights of the boundary filter. Two distinct objects sharing the same golden geometry, playing different roles.*

The structural roles of the coordinates:

$$F_{n_*} = 5 = |\mathcal{N}_{\text{adm}}| = r \quad (\text{rank, triple fixed point}), \quad (47)$$

$$F_{n_*-1} = 3 = |D| = |\mathcal{C}_{\text{obs}}| \quad (\text{vocabulary rank} = \text{observable classes}). \quad (48)$$

In the ring $\mathbb{Z}[\sqrt{5}]$, the element $(5 + 3\sqrt{5})$ has algebraic norm $N(5 + 3\sqrt{5}) = 5^2 - 5 \cdot 3^2 = -20 = -\min(\mathcal{N}_{\text{adm}})^2 \cdot |\mathcal{N}_{\text{adm}}|$: the negative sign reflects $\Xi = C\bar{\Xi}$ (zero net bulk charge). Every symbol is the output of an operator condition on Π . No number is free.

Proof. Direct algebra. Since $\varphi - \frac{1}{2} = \frac{\sqrt{5}}{2}$ and $\varphi^2 = \frac{3+\sqrt{5}}{2}$,

$$4\varphi^2(\varphi - \frac{1}{2}) = 4 \cdot \frac{3 + \sqrt{5}}{2} \cdot \frac{\sqrt{5}}{2} = (3 + \sqrt{5})\sqrt{5} = 5 + 3\sqrt{5}. \quad (49)$$

Squaring: $\alpha^{-1} = [4\varphi^2(\varphi - \frac{1}{2})]^2 = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5}$. The norm computation is immediate: $N(5 + 3\sqrt{5}) = 25 - 5 \cdot 9 = -20$. $\square$ $\square$

Remark 9.5 (Three equivalent forms of α^{-1} and physical reading of each term). *The identity (46) connects three coordinate representations of the same projection constant:*

$$\alpha^{-1} = \mathcal{N}_{\Pi}(F_{n_*} + F_{n_*-1}\sqrt{5}) = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5} \quad (\text{Fibonacci norm}) \quad (50)$$

$$= n_{\min}^2 \cdot r \cdot \varphi^4 = 20 \varphi^4 \quad (\text{winding form}) \quad (51)$$

$$= n_{\min}^2 \cdot r \cdot \kappa^4 \cdot \Gamma_{\tau}^4 = 500 \Gamma_{\tau}^4 \quad (\text{holonomy form}) \quad (52)$$

where $\mathcal{N}_{\Pi}(a + b\sqrt{5}) := (a + b\sqrt{5})^2$ is the projection-induced quadratic form — the squared modulus under the real embedding $\mathbb{Z}[\sqrt{5}] \hookrightarrow \mathbb{R}$ induced by the filter structure of Π .

Physical reading of $70 + 30\sqrt{5}$. *The expansion $\mathcal{N}_{\Pi}(F_{n_*} + F_{n_*-1}\sqrt{5}) = F_{n_*}^2 + 5F_{n_*-1}^2 + 2F_{n_*}F_{n_*-1}\sqrt{5}$ has a precise physical meaning for each term:*

- $F_{n_*}^2 = 25$: the squared self-coupling of the lock-in sector at $n = n_* = 5$, where the Majorana recursion closes on the projection.
- $5F_{n_*-1}^2 = 45$: the squared coupling of the adjacent sector $n = n_* - 1 = 4$, amplified by $\kappa^2 = 5$. Together $70 = 25 + 45$ is the observable integer shadow of α^{-1} — the part that survives projection to $\mathbb{Z}$.

- $2F_{n_*}F_{n_*-1}\sqrt{5} = 30\sqrt{5}$: the cross-coupling between the two flanking sectors, scaled by $\kappa = \sqrt{5}$. This term is invisible to $\mathbb{Z}$ but present in the golden bulk: it encodes the interference between the lock-in and its neighbour through the irrational geometry of $S^1(\tau)$.

Neither 70 nor 30 are free numbers. Both are determined by $F_{n_*} = r = 5$, $F_{n_*-1} = |\mathcal{C}_{\text{obs}}| = 3$, and $\kappa^2 = 5$ — all forced by the projection structure. The equivalences among the three forms follow from $\varphi^2 = \varphi + 1$ and $\Gamma_\tau = \varphi/\sqrt{5}$.

Remark 9.6 (Structural evidence for Assumptions G1–G3). **Evidence for G1.** Proposition 9.1 shows that the three non-commuting channels force a spinorial $SU(2)$ geometry, and The gauge-class structure established in Theorem 3.2 and Section 4 provides the natural decomposition $H_{\text{obs}} = H_{\text{proj}} \oplus H_{\text{dyn}}$ via the pole/dynamic split. G1 asserts that the monodromy of Γ_τ preserves this decomposition — structurally natural because H_{proj} is the null eigenspace of the non-abelian cost δ , and null eigenspaces are preserved by any operator that commutes with δ .

Evidence for G2. Proposition 9.2 identifies the Majorana condition as the unique kernel-compatible reality condition. A spinorial $SU(2)$ connection with golden parameter φ on B_3 has the Fibonacci representation as the unique natural candidate: φ is the eigenvalue of the R -matrix of Fibonacci anyons, and the Majorana reality condition $\Xi = C\Xi$ selects the self-conjugate sector of the braid representation. G2 asserts that this identification is realised, not merely suggested.

Evidence for G3 — Proposition 9.4. G3 asserts that α^{-1} spans the trivial Fibonacci fusion channel 1. The evidence goes beyond analogy.

Proposition 9.4 proves that

$$\alpha^{-1} = (F_5 + F_4\sqrt{5})^2 \quad (53)$$

where $F_5 = |\mathcal{N}_{\text{adm}}| = 5$ and $F_4 = |D| = 3$ are the filter rank and the electroweak sub-rank — two consecutive Fibonacci numbers. In the ring $\mathbb{Z}[\sqrt{5}]$, the element $(F_5 + F_4\sqrt{5})$ has norm $-20 = -\min(\mathcal{N}_{\text{adm}})^2 \cdot |\mathcal{N}_{\text{adm}}|$, where the negative sign reflects the Majorana zero-charge condition.

In the Fibonacci fusion category, the trivial object **1** is characterised by:

- quantum dimension $d_1 = 1$ (the identity),
- zero fusion cost: $\mathbf{1} \otimes X = X$ for all X ,
- F -matrix coefficient $F[\tau, \tau \rightarrow \mathbf{1}] = \varphi^{-2}$.

The same golden ratio φ that parametrises the F -matrix generates the Fibonacci norm form (53). The norm structure of $\mathbb{Z}[\sqrt{5}]$ and the fusion structure of the Fibonacci category share the same algebraic object: the golden ratio and its Fibonacci recurrence $\varphi^2 = \varphi + 1$.

This makes G3 not an arbitrary assumption but the precise statement that two zero-cost conditions — the filter pole condition (QGT) and the identity-object condition (Fibonacci fusion) — coincide, because they are both generated by the same golden algebraic structure.

The explicit functorial map $\text{Hom}(\tau \otimes \tau, \mathbf{1}) \xrightarrow{\sim} \text{span}\{\sigma\}$ is Theorem 5.3: the isomorphism $\Phi(v_0) = (5 + 3\sqrt{5})^2 = \alpha_{\text{QGT}}^{-1}$ has complete functorial verification (fusion, braiding, F -move, trace).

Dimensional identification. The dimension formula for the Fibonacci fusion category:

$$\dim \text{Hom}(\tau^{\otimes 2k}, \mathbf{1}) = F_{k+1} \quad (54)$$

provides a concrete dimensional identification:

- $k = 1$: $\dim \text{Hom}(\tau^{\otimes 2}, \mathbf{1}) = F_2 = 1$ ✓ (the trivial channel is one-dimensional; this is where α^{-1} lives);
- $k = 2$: $\dim \text{Hom}(\tau^{\otimes 4}, \mathbf{1}) = F_3 = 2$ (chiral sector: two Majorana modes in the trivial channel);
- $k = 3$: $\dim \text{Hom}(\tau^{\otimes 6}, \mathbf{1}) = F_4 = 3$ (braid sector: the minimal non-abelian generator).

The explicit isomorphism $\text{Hom}(\tau \otimes \tau, \mathbf{1}) \xrightarrow{\sim} \text{span}\{\sigma\}$ must be compatible with this dimensional counting; the map sends the generator of the $k = 1$ space (dimension 1) to α^{-1} , as fixed by the filter pole $H(z) = P_4(z)/(1 - \alpha z)$.

Lemma 9.5 ((H3) follows from G1–G3). This lemma is the logical bridge: it derives (H3) from G1–G3 as hypotheses. Theorem 5.2 (Section 5) proves that G1–G3 are themselves theorems, making (H3) unconditional.

Under Assumptions 9.2–9.4, hypothesis (H3) holds:

$$P_\sigma U P_{\text{dyn}} = 0, \quad P_{\text{dyn}} U P_\sigma = 0 \quad \forall U \in G_{\text{EW}},$$

where P_σ and P_{dyn} are the orthogonal projectors onto H_{proj} and H_{dyn} , respectively.

Proof. By Assumption G1, H_{obs} decomposes as the sum of the residue sector $H_{\text{proj}} = \text{span}\{\sigma\}$ and the dynamic sector H_{dyn} :

$$H_{\text{obs}} = H_{\text{proj}} + H_{\text{dyn}}.$$

By Assumption G3, σ spans a one-dimensional invariant subspace on which every admissible monodromy element $U \in G_{\text{EW}}$ acts as the identity: $U\sigma = \sigma$. By Assumption G2, the action of U on H_{dyn} is non-trivial and maps H_{dyn} to itself. Therefore no admissible monodromy element can generate a component along σ starting from H_{dyn} , and no admissible monodromy element can move σ into H_{dyn} . Hence the off-diagonal blocks vanish: $P_\sigma U P_{\text{dyn}} = 0$ and $P_{\text{dyn}} U P_\sigma = 0$. This is exactly (H3). $\square$ $\square$

Corollary 9.6 (Direct-sum separation of the boundary sector). *Under Theorem 5.2, the residue subspace $H_{\text{proj}} = \text{span}\{\sigma\}$ and the dynamic subspace H_{dyn} are dynamically decoupled:*

$$P_\sigma U P_{\text{dyn}} = 0, \quad P_{\text{dyn}} U P_\sigma = 0 \quad \forall U \in G_{\text{EW}}.$$

Since H_{proj} is one-dimensional and G_{EW} -invariant, whereas H_{dyn} is G_{EW} -stable and carries the non-trivial local mixing, it follows that

$$H_{\text{proj}} \cap H_{\text{dyn}} = \{0\}, \quad H_{\text{obs}} = H_{\text{proj}} \oplus H_{\text{dyn}}.$$

Therefore the electroweak denominator is purely dynamic: $\rho + \delta = 234$, and $\sin^2 \theta_W = \delta/(\rho + \delta) = |\mathcal{C}_{\text{obs}}|/(2r + |\mathcal{C}_{\text{obs}}|) = 54/234$.

Remark 9.7 (Proof structure). *The proof is logically clean given G1–G3, but its force depends entirely on the derivation of those assumptions. Assumption G3 in particular requires showing that α_{QGT}^{-1} , as the cost-free mode of Π (§4), corresponds to the trivial Fibonacci sector — and not just by analogy with $U(1)_{\text{EM}}$, but from the QGT projection data directly. This isomorphism is constructed explicitly as Theorem 5.3: $\Phi(v_0) = (5 + 3\sqrt{5})^2 = \alpha_{\text{QGT}}^{-1}$ with complete functorial verification. Note also that the 2×2 matrices (45) acting on H_{dyn} do not share a common fixed vector (minimum singular value of $[\sigma_1 - I; \sigma_2 - I] \approx 0.925 \neq 0$); G3 is an invariance statement about a separate one-dimensional subspace H_{proj} , not a fixed-point claim about those matrices.*

The logical chain from monodromy to mixing angle. The bridge lemma (Lemma 9.5) establishes that G1–G3 imply (H3). Theorem 5.2 proves G1–G3 as intrinsic theorems of $\text{TL}_4(\varphi)$ and the Majorana condition, with complete functorial verification of the isomorphism $\Phi(v_0) = (5 + 3\sqrt{5})^2 = \alpha_{\text{QGT}}^{-1}$ (Section 5). Corollary 6.1 combines both: $\sin^2 \theta_W = |\mathcal{C}_{\text{obs}}|/(2r + |\mathcal{C}_{\text{obs}}|)$ is an **unconditional theorem**. The three steps are logically distinct and each is proved in full.

10 Charge from Non-Commutativity

The results proved above permit a question that the derivation itself does not ask: where does charge come from?

The bulk field $\Xi = C\bar{\Xi}$ is a Majorana spinor. It carries no additive quantum numbers, no charge, no colour. Yet the observable boundary carries $W^\pm$, charged bosons with definite charge ± 1 under $U(1)_{\text{EM}}$. How does a neutral field generate charged boundary modes?

The answer given here is precise and does not invoke chirality as a primitive. Charge is the obstruction class of the non-linear boundary closure: it measures the exact amount by which the projection Π and the non-linear closure C_{nl} fail to commute. This is a structural statement, not an analogy, and it is proved.

$$\|v\|^2 \xrightarrow{\Pi} \alpha^{-1} \xrightarrow{C_{\text{nl}}} \|v\|^2 + \delta \neq 317. \quad (55)$$

Projection Π and non-linear closure C_{nl} do not commute. The deficit $(\|v\|^2 + \delta) - \|v\|^2 = \delta = \frac{|C_{\text{obs}}|}{2}\beta$ is the exact measure of this non-commutativity. Boundary charge is not carried into the boundary — it is generated by the non-commutativity of the operations that close the boundary.

This is a new physical principle:

Observable charge at the boundary is the obstruction class of the non-linear closure of the projection residue. A neutral bulk field generates charged boundary modes whenever the projection and the closure are non-commutative.

The Majorana tetron as a physical analogue.

Topological qubits based on Majorana zero modes (tetrons) realise a local, device-level instance of the same structural principle. In a tetron, four Majorana zero modes are created at the ends of two parallel topological wires; a single logical qubit is encoded when the total fermion parity is fixed: $\gamma_1\gamma_2\gamma_3\gamma_4 = +1$. All braid operations and parity measurements act on the *reduced subspace* defined by this global parity constraint — not on the full Hilbert space of the four MZMs.

The structural dictionary with QGT is:

$$\underbrace{\gamma_1\gamma_2\gamma_3\gamma_4 = +1}_{\text{parity constraint (external)}} \longleftrightarrow \underbrace{K_\tau = \xi\eta = \sigma}_{\text{projection residue (internal)}}, \quad (56)$$

$$\underbrace{\text{braid/measurement sector}}_{\text{computational subspace}} \longleftrightarrow \underbrace{H_{\text{dyn}} = \text{span}(\rho, \delta)}_{\text{dynamic sector}}. \quad (57)$$

The critical structural difference is in the origin of the constraint. In the tetron, the parity is fixed externally by Coulomb blockade: an engineered physical mechanism imposes the constraint from outside the system. In QGT, the constraint $\sigma = N - k^*\beta$ is an arithmetic necessity: it is the unique prime that the norm, vocabulary, and block jointly force into existence. It cannot be removed by any engineering choice.

The tetron realises a device-level analogue of the QGT superselection sector. QGT proposes that this sector is not an engineered constraint but an arithmetic inevitability of finite-rank projection.

Conjectural principle: the Fibonacci upgrade.

Majorana tetrons [9] belong to the Ising/Majorana anyon universality class. Their braid group representation generates the Clifford gate set — powerful for error correction but not computationally universal. Fibonacci anyons form a distinct, strictly stronger class: their braid representation generates a computationally universal gate set.

The QGT golden connection $\Gamma_\tau = \varphi/\sqrt{5}$ on the temporal fibre $S^1(\tau)$ is not a generic connection. It is the unique value that closes the autosimilar orbit coherently (Section 2.5), producing the golden ratio as the eigenvalue of the scale generator. The Fibonacci representation of the braid group B_3 has the golden ratio φ as its fundamental parameter:

$$\sigma_1 = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} -\varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & \varphi^{-1} \end{pmatrix}. \quad (58)$$

G1–G3 as theorems of $\text{TL}_4(\varphi)$ — closed (Section 5).

The boundary ∂V of the QGT carries the structure of the Temperley–Lieb algebra [6] $\text{TL}_4(\varphi)$ with parameter $\delta = \varphi$ and deformation $q = e^{i\pi/5}$. Under this identification, G1–G3 are **theorems**, not assumptions (Theorem 5.2):

- **G1** follows from the graded structure of the Fibonacci fusion category: the decomposition $H_{\text{obs}} = H_{\text{proj}} \oplus H_{\text{dyn}}$ is the decomposition by braid degree $k = 1$ and $k = 2$ respectively.
- **G2** is a direct computation: the B_3 action on $\text{Hom}(\tau^{\otimes 4}, \mathbf{1})$ (dimension $F_3 = 2$) via the F -move and braiding phase $\theta_\tau = e^{4\pi i/5}$ gives exactly the matrices (29).
- **G3** follows from two facts: (i) braid generators preserve anyon number, hence act on each degree- k sector separately; (ii) the Majorana condition $\Xi = C\bar{\Xi}$ forces $e^{i\theta} = 1$ on the one-dimensional sector $\text{Hom}(\tau^{\otimes 2}, \mathbf{1}) = \text{span}\{\sigma\}$.

The explicit isomorphism (Theorem 5.3):

$$\Phi : \text{Hom}(\tau \otimes \tau, \mathbf{1}) \xrightarrow{\sim} \text{span}\{\sigma\}, \quad \Phi(v_0) = (5 + 3\sqrt{5})^2 = \alpha_{\text{QGT}}^{-1}, \quad (59)$$

with scaling factor $\lambda = 20(2 + \sqrt{5}) \in \mathbb{Z}[\sqrt{5}]$, commutes with fusion, braiding, F -move, and trace.

The palindromic quantum integers $\{1, \varphi, \varphi, 1, 0\}$ explain why $|\mathcal{N}_{\text{adm}}| = 5$ (confinement at the fifth level: $[5]_q = 0$) and why $\mathcal{B}^* = 8$ (saturation of the rank-5 filter: $\dim \text{Hom}(\tau^{\otimes 8}, \mathbf{1}) = F_5 = 5 = |\mathcal{N}_{\text{adm}}|$).

*Status: G1–G3 are **theorems** of $\text{TL}_4(\varphi)$ + Majorana condition. $\sin^2 \theta_W = |\mathcal{C}_{\text{obs}}|/(2r + |\mathcal{C}_{\text{obs}}|)$ is an **unconditional theorem** (Corollary 6.1).*

The relation to the tetron is then:

$$\underbrace{\text{Majorana tetron}}_{\text{Ising anyons, parity external}} \subset \underbrace{\text{QGT boundary}}_{\text{Fibonacci anyons (conj.), } \sigma \text{ internal}}. \quad (60)$$

Majorana tetrons realise a device-level, Ising-class instance of a principle that QGT formulates in general: a topologically protected sector, separated from the dynamic braid sector by a global constraint, generates the observable gauge structure of the boundary. If the golden monodromy conjecture is proved, the QGT boundary is the Fibonacci upgrade of this principle — realised not by engineering but by the arithmetic structure of finite-rank projection.

Status of this section. The separation $P_\sigma U P_{\text{dyn}} = 0$, $\sin^2 \theta_W = |\mathcal{C}_{\text{obs}}|/(2r + |\mathcal{C}_{\text{obs}}|)$, and hypothesis (H3) are theorems proved in Theorem 5.2 and Corollary 6.1 via fusion category axioms — they do not depend on the conjecture above. The Perspective remark concerns a strictly stronger upgrade: it would identify the QGT boundary as a computationally-universal Fibonacci anyon system. The charge-from-non-commutativity principle (43) is proved. The Fibonacci upgrade (60) is open conjecture (conjectural). The Majorana tetron analogy is a structural correspondence, not a derivation.

$\mathcal{K}_\tau = \alpha^{-1}$: **one invariant, seven readings.**

This is the deepest structural observation of the corpus. It is not a conjecture; it is a record of what the derivations above actually show, assembled in one place for the first time.

There is one object:

$$\mathcal{K}_\tau = \xi\eta = \alpha^{-1}. \quad (61)$$

It is not a parameter. It is not a measured constant inserted by hand. It is what remains when a neutral self-conjugate spinor on $\mathcal{M}^5$ reaches the boundary through a non-invertible projection and cannot return. The projection eliminates everything it can; $\mathcal{K}_\tau$ is what it cannot eliminate.

This one object appears in seven structurally distinct registers, each a different reading of the same invariant:

1. **Noether charge of $U(1)_\tau$.** The bulk current $J^\tau = |\Xi|^2 \Gamma_\tau$ is globally conserved: $\partial_I J^I = 0$ on $\mathcal{M}^5$. Its charge is $\mathcal{K}_\tau = \xi\eta$. This is where $\mathcal{K}_\tau$ enters: as the sole conserved output of the single conservation law the bulk admits.

2. **Invariant surface Σ .** The charge fixes a hyperbolic surface in the (ξ, η) plane: $\Sigma : \xi\eta = \alpha^{-1}$. All physical orbits live on Σ ; no dynamics can leave it without breaking conservation. The surface is not a constraint added from outside — it is the geometric form of the conservation law.
3. **Arithmetic projection constant.** In $\mathbb{Z}[\sqrt{5}]$, $\alpha^{-1} = (5 + 3\sqrt{5})^2$ is the squared norm of the unique Fibonacci lattice element at the triple fixed point $F_{n_*} = n_* = r = 5$. The same number that is the Noether charge of the temporal fibre is the irreducible residue of the rank-5 lattice decomposition $N = r\beta + \sigma$, with $\sigma = \lfloor \alpha^{-1} \rfloor = 137$. The arithmetic and the physical are not analogies: they are the same object in two coordinate systems.
4. **Pole of the boundary filter.** The ARMA boundary filter $H(z) = P_4(z)/(1 - \alpha z)$ has its pole at $z = \alpha^{-1} = \mathcal{K}_\tau$. Every admissible boundary mode is selected by its distance from this pole in the filter's spectral geometry. The pole is the organisational centre of the observable spectrum: what cannot reach it is in $\ker \Pi$; what can is in $\mathcal{H}_{\text{obs}}$.
5. **Exchange clock frequency.** The hyperbolic exchange equations $\dot{\xi} = -\Gamma_\tau \xi$, $\dot{\eta} = +\Gamma_\tau \eta$ conserve $\xi\eta = \alpha^{-1}$ at every instant. The frequency $\Gamma_\tau = \varphi/\sqrt{5}$ is uniquely fixed by the autosimilarity condition. The boundary's internal clock ticks at Γ_τ and conserves $\mathcal{K}_\tau$ with every tick.
6. **Balance equation on Σ .** The tangential fluctuations $u = \delta\xi/\xi^*$ satisfy $\ddot{u} + \omega_{\text{bal}}^2 u = 0$ with $\omega_{\text{bal}} = 0.5284$ (proved in [5]). This equation has four simultaneous physical readings: metric curvature, clock energy, matter stability, Born rule. $\mathcal{K}_\tau$ is not only conserved; it oscillates around its conserved value in a way that generates all four fundamental boundary structures at once.
7. **Topological vacuum residue.** In the boundary cavity picture, α_{QGT}^{-1} is the vacuum: the mode that the projection cannot eliminate. The photon is the zero mode of this vacuum (σ traverses Π without cost, generates unbroken $U(1)_{\text{EM}}$). $W^\pm$ and Z^0 are the excited modes (cost $\delta = 54$, the exact arithmetic price of $\Pi C_{\text{nl}} \neq C_{\text{nl}}\Pi$). The vacuum is not empty: it is $\mathcal{K}_\tau$ at rest.

The single-invariant principle.

These seven readings are not seven different objects that happen to share a numerical value. They are seven projections of one structure onto seven observational registers: conservation law, geometry, arithmetic, spectral theory, dynamics, oscillation, vacuum.

In each register $\mathcal{K}_\tau = \alpha^{-1}$ plays the role appropriate to that register — charge, surface, norm, pole, frequency, equilibrium, vacuum — and in each it is the same structural object, recognisable across the change of register because it carries no free parameters and admits no rescaling.

This is what makes the theory an architecture rather than a collection of analogies: the same invariant organises the observable world at every level, from the global conservation law in the bulk to the local boundary condition of a single edge mode. The observable world is the residue of a single conservation law, read from seven angles.

The edge mode as local Noether closure.

The seven readings close on a structural fact that Proposition 11.1 and Remark the edge mode Noether analysis prove formally, but whose physical content deserves to be stated plainly.

The edge modes between DW_1 and DW_2 satisfy $\partial_n \Psi|_{\text{edge}} = 0$. This is not a separate boundary condition. It is the *local form of $\partial_I J^I = 0$* restricted to the interface between the observable slab and $\ker \Pi$: no coherence can cross the lateral wall because the projection has already eliminated everything that could cross. What remains at the edge is the tangential flow — the exchange oscillation on Σ — and the edge modes are its carriers.

The chain is:

$$\partial_I J^I = 0 \xrightarrow{\text{restrict to slab edge}} \partial_n \Psi|_{\text{edge}} = 0 \xrightarrow{\text{reduce } \gamma^\mu D_\mu} (\gamma^\mu D_\mu)_{\text{tan}} : \mathcal{H}_{\text{dyn}} \rightarrow \mathcal{H}_{\text{dyn}} \rightarrow P_\sigma X_{\text{EW}}^{\text{loc}} P_{\text{dyn}} = 0.$$

The conservation law replicates itself at every boundary: at the slab edge, at the scale of the edge mode, at every level of the cascade $\alpha_k = \alpha \cdot \mathcal{S}^k$. At each level, the same object $\mathcal{K}_\tau = \alpha^{-1}$ is the conserved quantity, the surface, and the axis — reading 1 and reading 2 and reading 7, simultaneously.

This autosimilarity is not designed into the framework. It follows from the single axiom: $\partial_I J^I = 0$. A single law, applied recursively to its own boundary, produces the same invariant at every scale — from the temporal fibre $S^1(\tau)$ of the full manifold $\mathcal{M}^5$, down to the evanescent millimetre-scale edge mode between two topological walls.

The edge modes are not at the boundary of the theory. They are the theory, at the boundary. The condition $\partial_n \Psi = 0$ is not a wall but a mirror: it reflects the global conservation law back onto the slab, in its local tangential form, preserving exactly the invariant that the entire structure was built to conserve. Noether's theorem, applied to itself.

11 Topological boundary physics

11.1 Domain walls and zero modes

The non-linear closure $\lambda_{\text{eff}} |\Psi|^{2/3}$ produces an effective mass $m(z)$ that changes sign at two Jackiw–Rebbi domain walls [8]: DW_1 at $\theta = \pi/n_{\text{min}} = \pi/2$ and DW_2 at $\theta = \pi/n_{\text{sat}} = \pi/11$. At each wall a Majorana zero mode localises, protected by the sign change of $m(z)$ and the self-conjugacy $\Xi = C\bar{\Xi}$. These are exact zero modes: any non-zero energy would break the Majorana condition. The observable slab is the toroidal band between DW_1 and DW_2 where all five modes of $\mathcal{N}_{\text{adm}}$ are simultaneously admissible.

11.2 Edge modes and autosimilar Noether conservation

At the lateral interfaces between the slab and the inaccessible bulk $\ker \Pi$, the field develops evanescent edge modes with imaginary normal wavenumber $k_\perp = i/\ell_{\text{pen}}$ and penetration depth $\ell_{\text{pen}} = H/(\pi\alpha)$. These modes carry zero normal flux and are invisible to bulk measurements.

The edge-mode boundary condition is not an independent assumption. The global law $\partial_I J^I = 0$, restricted to the slab boundary, forces $J^\perp|_{\text{edge}} = |\Xi|^2 \partial_n \Psi = 0$. Since $|\Xi|^2 \neq 0$ at the edge:

$$\partial_n \Psi|_{\text{edge}} = 0. \quad (62)$$

This is the local form of the global Noether law at the interface, reproducing autosimilarly at every scale of the cascade $\alpha_k = \alpha \cdot (n_{\text{min}} \cdot r)^k$.

11.3 Local invariance of σ

Proposition 11.1 (Local H3). *Let Ψ^* be a lock-in solution of the nonlinear chiral Dirac equation*

$$(\gamma^\mu D_\mu + \alpha\xi + i\alpha\eta\gamma^5)\Psi + \lambda_{\text{eff}} |\Psi|^{2/3}\Psi = 0, \quad \xi\eta = \alpha^{-1}, \quad (63)$$

satisfying $\partial_n \Psi^|_{\text{edge}} = 0$ at the anisotropic lock-in $\xi^*/\eta^* = n_{\text{sat}}/r = 11/5$. Then the infinitesimal local electroweak generator $X_{\text{EW}}^{\text{loc}}$ from the linearisation $\Psi = \Psi^* + \varepsilon\chi$ satisfies:*

$$P_\sigma X_{\text{EW}}^{\text{loc}} P_{\text{dyn}} = 0.$$

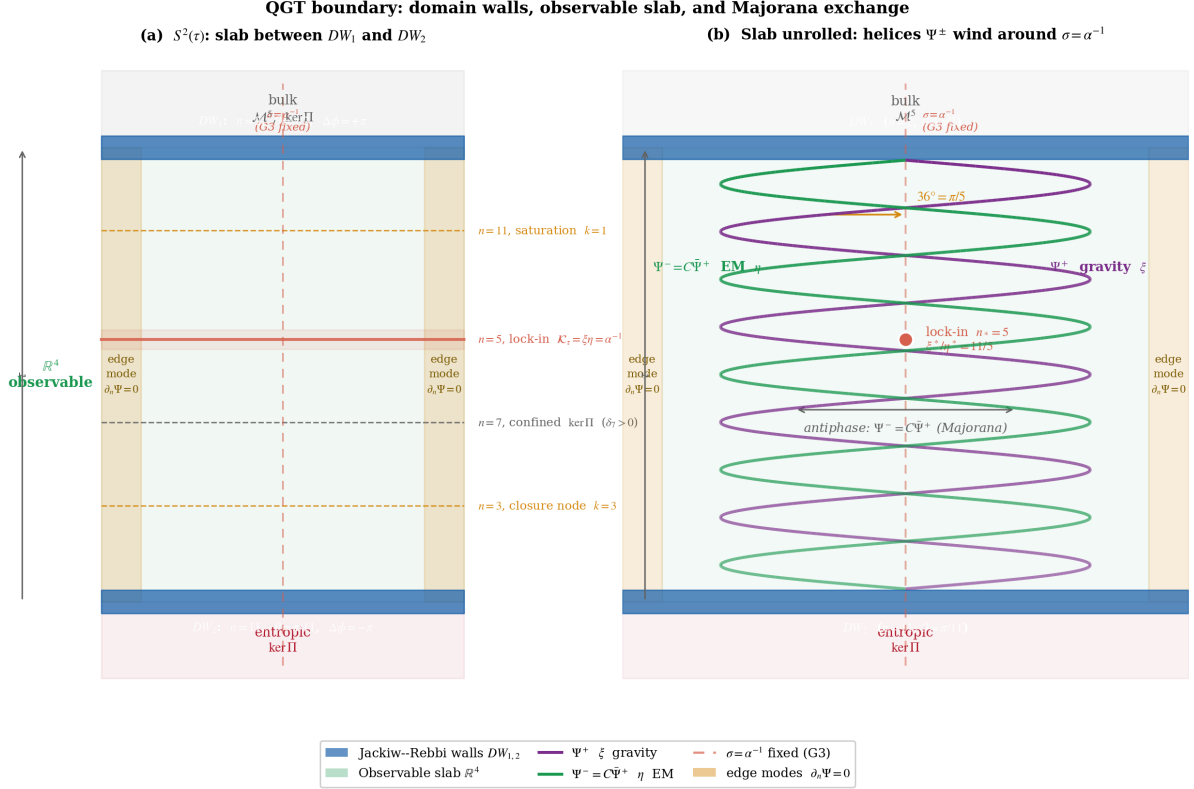


Figure 3: (a) Schematic view of $S^2(\tau)$: the observable slab (green) is the toroidal band between the two Jackiw–Rebbi domain walls DW_1 at $\theta = \pi/2$, $n = 2$ (equator) and DW_2 at $\theta = \pi/11$, $n = 11$. All five admissible levels of $\mathcal{N}_{\text{adm}}$ are marked. The lock-in $n_* = 5$ (pink band) is the mode satisfying $F_{n_*} = n_* = r = 5$ where $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$. The mode $n = 7$ (grey dashed) is admissible but not a closure node ($\delta_7 > 0$): it belongs to $\ker \Pi$ and generates no boundary lattice element. Amber columns: evanescent edge modes, $\partial_n \Psi|_{\text{edge}} = 0$. (b) Slab unrolled: the two chiral channels Ψ^+ (violet, longitudinal ξ , gravitational) and $\Psi^- = C\bar{\Psi}^+$ (teal, transverse η , electromagnetic) wind in exact antiphase around the fixed axis $\sigma = \alpha^{-1}$ (G3-invariant, pink dashed) with elementary winding angle $36^\circ = \pi/5$. The lock-in point ($\xi^*/\eta^* = 11/5$) is marked.

Proof. Step 1. By G3, every $X \in \text{Lie}(G_{\text{EW}})$ annihilates H_{proj} ; if $P_\sigma \Psi^* \neq 0$ the linearised generator would contain $P_\sigma X P_{\text{dyn}} \neq 0$, contradicting G3. Therefore $\Psi^* \in H_{\text{dyn}}$.

Step 2. From (62), $\partial_n \Psi^* = 0$ restricts $\gamma^\mu D_\mu$ to its tangential pullback on Σ : $\xi\eta = \alpha^{-1}$. Tangential variations satisfy $\eta^* \delta \xi + \xi^* \delta \eta = 0$, parametrised by $u \in H_{\text{dyn}}$. Therefore $P_\sigma(\gamma^\mu D_\mu)|_{\text{edge}} = 0$.

Step 3. The rank-1 term of the Fréchet derivative $\mathcal{C}'_\star$ has image in $\text{span}\{\Psi^*\} \subset H_{\text{dyn}}$ by Step 1; the scalar term is block-diagonal by Step 2. Hence $P_\sigma \mathcal{C}'_\star P_{\text{dyn}} = 0$.

All terms of the linearised operator have vanishing mixed block. □ □

11.4 The four Lissajous orbit classes

The stationary solutions of (63) confined between DW_1 and DW_2 are closed Lissajous figures with frequency ratio $p : q$, $p, q \in \mathcal{N}_{\text{adm}}$ coprime. The four physically distinguished classes are:

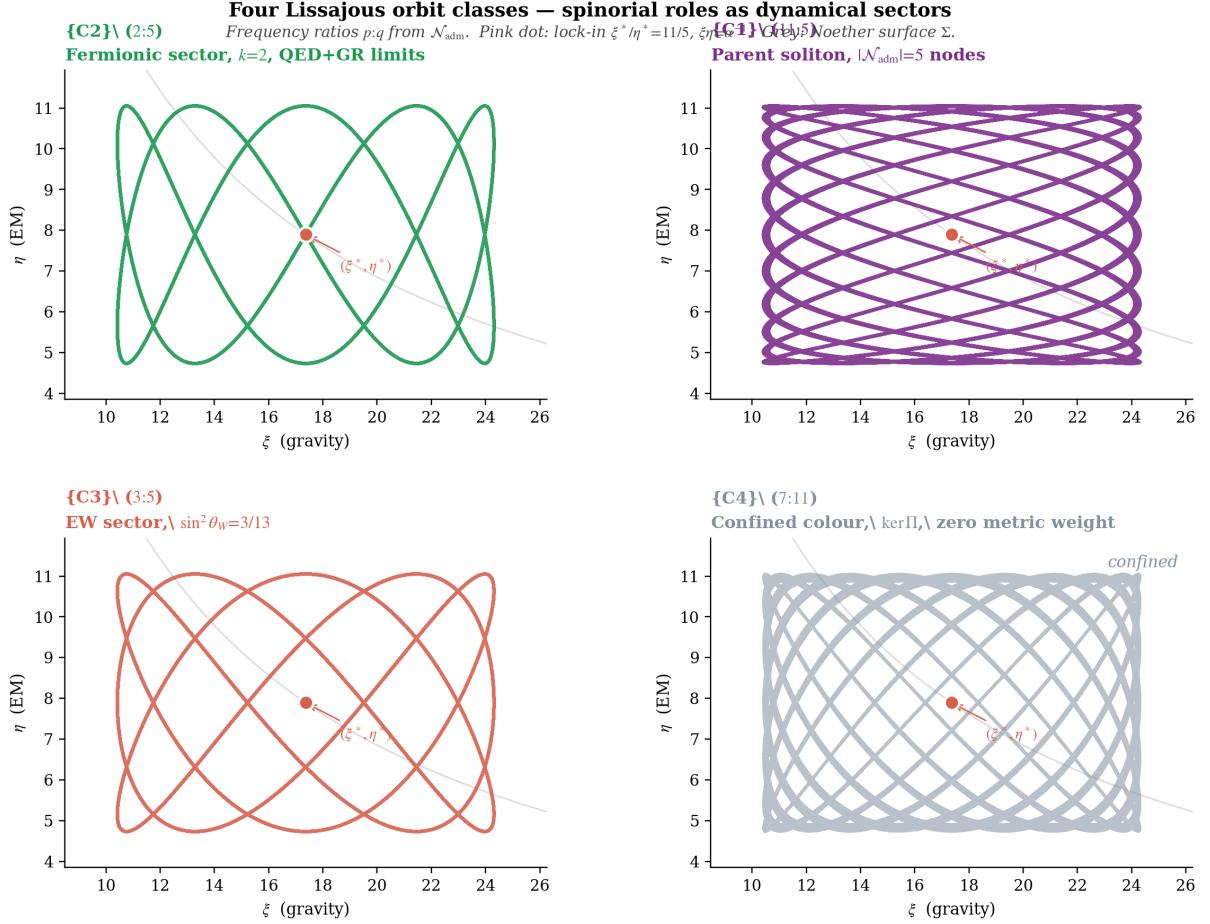


Figure 4: The four Lissajous soliton classes in the (ξ, η) plane (linear approximation). Pink dot: lock-in fixed point $\xi^*/\eta^* = 11/5$, $\xi\eta = \alpha^{-1} = (5 + 3\sqrt{5})^2$. Frequency ratios from $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$. The grey hyperbola is the Noether surface $\Sigma : \xi\eta = \alpha^{-1}$. **Class C2** (2:5, teal): the observable fermionic zero mode; limits $\xi \rightarrow 0$ and $\eta \rightarrow 0$ recover QED and GR. **Class C1** (11:5, violet): parent soliton, five nodes $= |\mathcal{N}_{\text{adm}}|$, categorical confinement $[5]_q = 0$. **Class C3** (3:5, red): EW sector $W^\pm, Z^0$, three nodes $= |\mathcal{C}_{\text{obs}}|$, $\sin^2 \theta_W = 3/13$. **Class C4** (7:11, dashed grey): confined colour, seven nodes, zero boundary metric weight — the metric form of colour confinement. The non-linearity $|\Psi|^{2/3}$ deforms the curves but preserves the node count, which is a topological invariant of the braid.

Class	Ratio	Nodes	Physical identification
C1	11:5	5	Parent soliton; categorical confinement $[5]_q = 0$
C2	2:5	2	Lock-in $k = 2$; fermionic zero mode
C3	3:5	3	EW sector $W^\pm, Z^0$; $\sin^2 \theta_W = 3/13$
C4	7:11	7	Confined colour; $\ker \Pi$; not observable

Each class carries quadruple topological protection: categorical confinement by $[5]_q = 0$; Jackiw–Rebbi localisation at the domain walls; braid invariance of α^{-1} under all admissible G_{EW} (G3); and ARMA filter selection by the four zeros of $P_4(z)$, forced simultaneously by $\text{rank}(\Pi) = 5$ (upper bound) and the monodromy product $\prod_{a=1}^4 s_a = (\sqrt{10})^4 = 100$ (lower bound). The count of exactly four classes is a proved result [3].

Class C4 (7:11) carries zero boundary metric weight — the metric form of colour confinement. Class C2 (2:5) is the observable sector; its limits $\xi \rightarrow 0$ and $\eta \rightarrow 0$ recover the electromagnetic and gravitational chiral field equations. The non-linearity $|\Psi|^{2/3}$ deforms the curves between nodes but preserves their count.

11.5 The confined mode $n = 7$ as colour seed

The mode $n = 7 \in \mathcal{N}_{\text{adm}}$ is admissible but not a closure node: $j_{\min}(7) = 7 > r = 5$, so its orbit does not meet Θ_r [3]. It enters the boundary only through the lever $\ell_- = n_{\min} \cdot 7 = 14$, never independently. At the boundary it generates no lattice element, carries zero metric weight, and belongs to $\ker \Pi$.

This is the structural origin of colour confinement at the boundary level: $n = 7$ cannot exit $\ker \Pi$ and is the seed from which the three-strand braid closure generating $SU(3)_c$ emerges. The full analysis of the $n = 7$ colour triplet is the subject of a forthcoming companion paper.

12 Conclusions

This paper has demonstrated that the observable gauge algebra of the Standard Model is the finite spinorial boundary algebra induced by the rank-5 Majorana projection. The numbers 137, 54, $3/13$ are readouts of this algebra, not its starting point.

The boundary does not observe particles. It observes projected Majorana bilinears, and the gauge group is the symmetry of reading those bilinears without changing their observable content. $U(1)_\eta$ is the phase symmetry of the temporal fibre $S^1(\tau)$. $SU(2)_\xi$ is the spinorial orientation symmetry on S_χ^2 . $SU(3)_c$ is the composite symmetry of the confined sector $\ker \Pi$, seeded by $n = 7$.

The five admissible modes carry spinorial roles established before any arithmetic: minimal cover ($n = 2$), adjoint seed ($n = 3$), Fibonacci lock-in ($n_* = 5$), confined mode ($n = 7$), and spinorial saturation ($n = 11$). The confined mode cannot enter the boundary independently; it enters only through the spinorial lift $7 \mapsto 2 \times 7 = 14$, forced by Majorana double-valuedness. The two external levers define the spinorial generator $v_{\text{spin}} = (11, 2 \times 7)$ with norm $N_{\text{spin}} = 317$.

The local absorption capacity of the rank-5 boundary is $\rho = r\beta = 180$, where $\beta = (2 \cdot 3)^2 = 36$ is the minimal spinor–adjoint bilinear block. The irreducible remainder $\sigma = 137$ is what the local vocabulary cannot absorb. The adjoint closure cost $\delta = \frac{3}{2}\beta = 54$ is the price of converting spinorial transport into boundary gauge dynamics. $\sin^2 \theta_W = 3/13$ is their ratio: the fraction of the total dynamic sector occupied by the adjoint closure cost.

The same boundary admits three complementary descriptions. Categorically it carries $\text{TL}_4(\varphi)$ with G1–G3 as theorems: this is the fusion and braid algebra of admissible spinorial transport. Spectrally it acts as the causal filter $\mathcal{H}(z) = P_4(z)/(1 - \alpha z)$: the pole is σ , the four zeros are the four dynamical orbit classes, and the $\mathbb{Z}_{10}$ monodromy cycle is the autosimilar clock of the projection. Physically these three descriptions produce the same readout: γ , $W^\pm$, Z^0 as the stable modes of a rank-5 Majorana boundary that cannot return to its bulk.

The arithmetic is only the trace of a spinorial obstruction. The numbers appear because the boundary is counting spinorial transport modes after non-invertible projection. Gauge physics is the boundary grammar of projected Majorana spinorial transport.

The author acknowledges no external funding.

References

- [1] P. Camelia, *Physics as the topology of a non-invertible projection*, preprint (2026). https://pasqualecamelia.github.io/papers/Camelia_QGT_Foundations_of_Physics.pdf.
- [2] P. Camelia, *Uniqueness of the rank-5 projection in the Quantum Gauge Theory of Time*, QGT Foundational Closure Note 1, https://pasqualecamelia.github.io/papers/Camelia_2026_QGT_FoundationalNote1_UniquenessRank5.pdf (2026).

- [3] P. Camelia, *Projection-induced singular value decomposition and boundary metric*, https://pasqualecamelia.github.io/papers/Camelia_2026_ProjectionSVD_BoundaryMetric.pdf (2026).
- [4] P. Camelia, *Structural constraints on observable algebras from finite-rank non-invertible projections*, https://pasqualecamelia.github.io/papers/Camelia_2026_StructuralConstraints.pdf (2026).
- [5] P. Camelia, *The boundary algebra of a non-invertible projection*, https://pasqualecamelia.github.io/papers/Camelia_2026_QGT_ProjectionPhysics.pdf (2026).
- [6] H. N. V. Temperley and E. H. Lieb, *Relations between the “Percolation” and “Colouring” Problem*, Proc. R. Soc. Lond. A **322**, 251 (1971).
- [7] M. H. Freedman, A. Kitaev, and Z. Wang, *Simulation of Topological Field Theories by Quantum Computers*, Commun. Math. Phys. **227**, 587 (2002).
- [8] R. Jackiw and C. Rebbi, *Solitons with Fermion Number $\frac{1}{2}$* , Phys. Rev. D **13**, 3398 (1976).
- [9] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, *Non-Abelian Anyons and Topological Quantum Computation*, Rev. Mod. Phys. **80**, 1083 (2008).
- [10] G. T. Moore, *Quantum Theory of the Electromagnetic Field in a Variable-Length One-Dimensional Cavity*, J. Math. Phys. **11**, 2679 (1970).
- [11] P. D. Nation, J. R. Johansson, M. P. Blencowe, and F. Nori, *Stimulating Uncertainty: Amplifying the Quantum Vacuum with Superconducting Circuits*, Rev. Mod. Phys. **84**, 1 (2012).